Project Number: 1

Project Title: AcademicHub

Project Clients: Will Midgley

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 2 group

Main contact: Will Midgley

Background:

Academics have lots of spinning plates to keep track of: HDR students; journal papers; conference papers and deadlines; reviewing of journal papers; and undergraduate students. They also have to keep track of their metrics for things like grant proposals and awards applications.

It would be great to have this all in one web interface that allows academics to keep all of the relevant information in one place.

Harzing's Publish or Perish is a frequently-used software tool to keep track of academic metrics.

Requirements and Scope:

Build a single piece of software that can bring together the various threads of an academic's life.

One (ideally online) portal should show an academic a snapshot of the status of the various parts of their role. It should be possible to drill down into details, and see visual representations of key metrics like citations and publications.

Required Knowledge and skills:

Requirements:

- Online portal

- Use API calls to extract information from databases like ORCiD and Web of Science

- Summarise current journal submissions and statuses

- Summarise in-progress journal/conference submissions

- Summarise students (this may have to be manual data-entry for the academic, unless you can organise access to the Uni's student information system using SSO)

Additional:

- Use Uni SSO

- Ingest emails and use them to update e.g. manuscript and reviewing statuses

- Automatically scrape conference deadlines

Expected outcomes/deliverables:

Codebase

Database schema

Documentation

User guide

Project Number: 2

Project Title: Automated Program Repair System Based on Large Language Models/Artificial Intelligence

Project Clients: Jiaojiao Jiang

Project specializations: Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Large language model;

Number of groups: 3 groups

Main contact: Jiaojiao Jiang

Background:

As the scale and complexity of software development continue to increase, fixing code defects and vulnerabilities has become an essential challenge for development teams. Traditional manual repair methods are not only time-consuming and labour-intensive but also prone to omissions, making it challenging to meet the needs of modern software development. In recent years, artificial intelligence technologies, huge language models (LLMs) and deep learning models have made code understanding and generation breakthroughs, providing new possibilities for automated program repair. Currently, the mainstream code repair methods in the industry are mainly categorized as follows: Pattern-matching-based methods: repair using predefined error patterns and repair templates, but the scope of the application is narrow. Test case-based approaches: program validation and repair generation using test cases, but relies on high-quality test suites. Machine learning-based approaches use deep learning models to learn code patterns and repair strategies, but early models have limited repair effectiveness and generalization capabilities. Large language model-based approaches utilize the robust semantic understanding and code generation capabilities of pre-trained language models, which show better potential for repair. In such a context, developing an automated project repair system based on advanced AI models has essential research value and practical significance.

Goal:

Project goals may include but are not limited to（not include all）:

Establishing a complete project remediation process

Integrate multiple AI/large language models/deep learning models for code understanding and analyze program dependencies.

Designing model fusion strategies

Developing model integration framework

Implement collaborative decision-making mechanisms between different models.

Designing dynamic model selection strategies

Optimize the comprehensive evaluation of multi-model outputs

Establish a multi-dimensional restoration quality assessment system

Realize automated testing and verification mechanism

Provide manual audit interface

Requirements and Scope:

This project aims to develop an automated project repair system based on various AI/big language/ models/deep learning/machine learning models to enable the automatic detection and repair of code defects. The automated program repair system must include the modules needed to accomplish the task, and each module must be able to fulfil its responsibilities correctly.

Required Knowledge and skills:

The system developed in this project needs to be able to perform automated program repair tasks. The system must input a buggy program and output code that fixes the bugs and passes validation. The system may need to include tasks such as defect detection of the code, dependency analysis of the code, and generation of program patches. Moreover, the system requires a reasonable test module to test whether the fixed code is fixed correctly or not.

Expected outcomes/deliverables:

Expected deliverables include:

An automated program repair system that is trained and capable of functioning and completing its tasks.

Detailed developer documentation/technical documentation/user's manuals and other documentation

Standards-compliant code comments/system demos/usage resources, etc.

Project Number: 3

Project Title: Enhancing Spatial Fidelity in Virtual Reality: Adaptive Viewpoint and Scene Scaling for Improved User Comfort

Project Clients: Dr Peter Wagner, Professor Juno Kim

Project specializations: Software Development;System/game Development;Human Computer Interaction (HCI);

Number of groups: 2 group

Main contact: Dr Peter Wagner, Professor Juno Kim

Background:

Head-mounted display (HMD) virtual reality (VR) has become an increasingly popular medium for immersive experiences, spanning entertainment, education, training, and rehabilitation. However, a significant proportion of users still experience discomfort that inhibits prolonged or repeated VR use. While much research has been dedicated to improving visual resolution and aligning visual simulations with postural positioning, an often-overlooked factor is the spatial alignment of virtual environments with real-world sizes and positions.

In interactive VR applications, spatial fidelity is crucial for haptic feedback and proprioceptive alignment. The perceived and actual positions of a user's body, limbs, and hands within the virtual environment must be accurately mapped to ensure a natural interaction. Unlike 2D environments, where size perception is predominantly influenced by visual cues, 3D environments require a careful calibration of both viewpoint projection and object scaling to match real-world user experiences. Misalignment can lead to discomfort, disorientation, and diminished presence, negatively impacting the overall VR experience.

Requirements and Scope:

- Develop a Unity VR application that enables real-time manipulation of object scaling and positioning within a scene.
- Implement a customizable viewpoint projection centre that aligns with users' natural eye positions and field of view.
- Conduct user testing to evaluate the effectiveness of these manipulations
- Provide an interface for users to fine-tune spatial settings accordingly

Required Knowledge and skills:

- Dynamic Spatial Manipulation: Enable real-time scaling and positioning adjustments of objects within the VR scene.
- Viewpoint Projection Centre Control: Implement adjustable viewpoint settings to match the user's natural visual perspective.
- User Calibration Interface: Develop an intuitive UI for users to fine-tune spatial settings based on their physiology.
- User Data Logging & Analysis: Implement a system for collecting user feedback and interaction data to refine spatial alignment settings.

Expected outcomes/deliverables:

- A fully-recompilable Unity VR project an necessary assets that allow dynamic spatial manipulation of objects and viewpoint adjustments.
- Recommendations for integrating spatial customization features into commercial VR applications.

Project Number: 4

Project Title: Optical Tracking Pipeline for UNSW C14 Telescope System

Project Clients: Yang Yang

Project specializations: Software Development;Big data Analytics and Visualization;Astronautical instrumentation and astrodynamics;

Number of groups: 1

Main contact: Yang Yang

Background:

To integrate and refine different modules of image processing and orbit determination for an automated pipeline for optical tracking of resident space objects by using UNSW's C14 telescope system.

Requirements and Scope:

With the growing number of satellites and debris orbiting the Earth, there is an urgent need for effective space tracking and controlling capabilities. However, current Space Situational Awareness (SSA) research lacks the capacity, capability, and diversity to fully track and identify objects with sufficient accuracy for space traffic management. To improve SSA observations and provide more accurate information about resident space objects (RSOs), the existing 14-inch Schmidt Cassegrain Telescope (C14) located on the Kensington Campus is being upgraded. Besides the hardware development, a software pipeline is also under development, including the target selection, telescope control, image processing and orbit determination. This project aims to refine an existing prototype software pipeline to work with the telescope for all model integration, testing and operation.

Required Knowledge and skills:

This project requires students to understand and test the satellite streak detection, astrometric positioning and orbit determination modules within the existing optical tracking software. The software package is written in Python and can be run via Docker. Students may need to develop additional code snippets, if necessary, under the guidance and instructions provided by the supervisor. The project also involves visiting the UNSW observatory multiple times to conduct rigorous testing and validate the software using satellite tracking data. The project will culminate in an observation campaign demonstration by operating the telescope using the software pipeline.

Expected outcomes/deliverables:

A refined optical tracking pipeline software package validated by using UNSW telescope data.

Project Number: 5

Project Title: Conversational Agent as facilitator for Group Projects

Project Clients: Sonit Singh, Basem Suleiman

Project specializations: Software Development;Artificial Intelligence (Machine/Deep Learning, NLP);Web Application Development;

Number of groups: 2 group

Main contact: Sonit Singh, Basem Suleiman

Background:

Most of the advanced courses at university level have team-based project work, where a team of four or five students, work towards delivering solution to a real-world problem. Group projects allow students to develop crucial collaborative skills, apply theoretical knowledge to solve real-world problem, gain exposure to diverse perspectives, practice time and project management, and learn how to resolve conflicts. Although team-based project work is a rewarding experience but it has many challenges, such as unequal contribution, poor communication, conflicts, poor leadership, unclear goals, uneven task distribution, and withholding information. The aim of this project is to design and develop a conversational agent that can enhance group discussions in a course-based group project work. This conversational agent can act as a facilitator in group discussions, taking in consideration of not only weekly group project criteria but also by managing time, encourage members to participate evenly, and organising member's diverse opinions.

Requirements and Scope:

The project scope is to design and develop a conversational agent using state-of-the-art Large Language Models (LLMs) to support group project.

Required Knowledge and skills:

TBD

Expected outcomes/deliverables:

Source code, user guide, report, presentation

Project Number: 6

Project Title: Design and development of a project recommendation system based on Profiles Complementarity

Project Clients: Sonit Singh, Basem Suleiman

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 2 group

Main contact: Sonit Singh, Basem Suleiman

Background:

Choosing a particular project and team members is a challenging feat and impacts team's success in delivering quality project. It involves a nuanced understanding of team members' skills, project requirements, and overall project goals. Traditional methods often rely on manual assessments and subjective decision making, leaving room for inefficiencies and mismatches. The goal of this project is to use data-driven and objective approach using state-of-the-art artificial intelligence technologies. The AI is used to create detailed skill profiles for each member by analysing their past project performances, certifications, and completed courses. The developed system also identifies skills required to deliver a particular project successfully. Based on individual profiles and required skills for the project, the recommender system ensures that the right team members are selected for the right project based on their right skillset.

Requirements and Scope:

The scope of the project is to develop a recommender system which can team members complementary skillset, project skills requirement, and recommending best matches.

Required Knowledge and skills:

TBD

Expected outcomes/deliverables:

Source code, user guide, report, and presentation

Project Number: 7

Project Title: CeleStE Conference Management Software

Project Clients: Arjun Radhakrishnan

Project specializations: Software Development;Web Application Development;Mobile Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Human Computer Interaction (HCI);

Number of groups: 5 groups

Main contact: Arjun Radhakrishnan

Background:

As an organiser of an international conference held in Sydney, you are to prepare an application that can enable the user as well as the administrator to have complete control over the whole program. The conference runs for 3 days (Agenda to be provided) which will include up to 2000 participants in person and 1000 virtually. The requirement is a digital platform that works well as a web page as well as a mobile application. The expectation is to have the users get an unforgettable experience , ability to network with others and participate in workshops and other events (that run parallelly to the main event) based on their interests providing ability to make last minute changes and chat with an admin person/ bot for help if need be.

The objective is

- Reducing manual efforts in organizing and running conferences.
- Enhancing attendee engagement through interactive features.
- Providing a scalable and secure platform for conference management.
- Offering robust customer service and support during events.
- Ensuring ease of access to schedules and relevant conference information.

Requirements and Scope:

To streamline the organization and management of conferences by providing a digital platform for handling agendas, participants, customer service chats, and other essential conference activities. The system will serve event organisers, speakers, attendees, and support staff to ensure a smooth experience before, during, and after the event.

Required Knowledge and skills:

- Agenda Management: Schedule creation, session tracking, and updates.

- Participant Management: Registration, profile creation, and role assignments (attendee, speaker, organizer, etc.).
- Communication & Networking: Chat functionality, direct messaging, and discussion forums.
- Customer Support Chat: AI-based and human-assisted chat support.
- Notifications & Alerts: Real-time event updates, reminders, and announcements.
- Speaker & Exhibitor Information: Profiles, session details, and promotional content.
- Ticketing & Access Control: Digital tickets, QR code scanning, and seat reservations.
- Feedback & Surveys: Post-event feedback collection and analysis.

Expected outcomes/deliverables:

- Users should be able to register, log in, and manage their profiles.
- Organizers should be able to create and update event agendas.
- Participants should have access to conference schedules and session details.
- A chat system should be integrated for networking and customer support.
- Push notifications should alert users about schedule changes and announcements.
- The system should support multiple roles with different permission levels.
- Data analytics and reporting tools should be available for post-event analysis.

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Project Number: 8

Project Title: System for finding key events or users in sentiment changes at macro or micro levels on social media

Project Clients: Daniel Kienast & Jiaojiao Jiang

Project specializations: Software Development;Computer Science and Algorithms;Big data Analytics and Visualization;Artificial Intelligence (Machine/Deep Learning, NLP);Web Application Development;

Number of groups: 3 groups

Main contact: Daniel Kienast & Jiaojiao Jiang

Background:

Current research for hate speech, extremist and other detection methods look at the individual post data. Not much research has been found at the use of whole user classification in this field. But something that comes from this is the ability to find what key events or users are the catalyst for this change in sentiment. Such events could be seen with the use of a hashtag a word or words. It would be helpful for researchers to be able to find out which events or users have become the catalyst for a change in sentiment or discourse. This should be

Requirements and Scope:

The scope of the project is to create software that can find these events in a social media dataset. This dataset can either be obtain by the students or a dataset can be provided. Sentiment analysis should be run on the data to find out the sentiment (can either be positive,negative or more such as positive, neutral, negative, extremists) of each post, so one can see any changes in sentiment. And look at the use of a hashtag/etc over time, then use a system to find its spread and rise through a network. After this a interface is need to see the results of the software in a visual easy to understand system. This should also allow for some customization such as given it a custom time range to search for. Or see in how many events a user is linked to.

Required Knowledge and skills:

System that can find these users or events

If they went with the NLP option then by how much did this event change the sentiment of a group and each user

A website or other visual display where one can search through all catalysts, and be able to change some parameters such as the time range to search for.

Expected outcomes/deliverables:

Source code

Guide on how to use it

Any interesting findings

Project Number: 9

Project Title: Enhancing Genetic Search for A Novel Fuel Cell Vehicle Routing Problem

Project Clients: Boyu Liu

Project specializations: Computer Science and Algorithms;

Number of groups: 3 groups

Main contact: Boyu Liu

Background:

The increasing emphasis on sustainable transportation has led to the development of Fuel Cell Vehicles (FCVs) as a viable alternative to conventional internal combustion engine vehicles. However, the integration of FCVs into transportation networks introduces a new set of challenges, particularly in routing optimization due to limited hydrogen refueling infrastructure, energy consumption variability, and operational constraints.

Vehicle Routing Problems (VRP) is a classic problem widely studied in the field of transportation engineering, but the VRP for FCVs (FVRP) has not been studied well. As I am pushing forward the mathematical modelling of FVRPs, I find the solving the problem has no good ways, as nonconvex optimization by commercial solvers run for an unacceptable long time and evolutionary algorithms have difficulties in obtaining precise optimal solutions. Therefore, the goal of this project is to combine the advantages of both sides and achieve better computational performance.

Requirements and Scope:

1. Problem Definition: What is a FVRP, what is the mathematical model of it.

2. Optimization Framework: How to use genetic algorithm to solve problem and how to improve the performance.

Required Knowledge and skills:

Overall this project is an easy task with moderate algorithm and mathematical knowledge required. The students are expected to complete the following tasks:

1. Understand the mathematical model of FVRP (will be provided)

2. Use genetic algorithm (or hybrid genetic search or other GA variants) to solve a FVRP as benchmark

3. Find a way to improve the performance of GA on solving FVRP (directions can be given, details needs to be explored by the students)

Codes has to be either Python or MATLAB.

Expected outcomes/deliverables:

1. Source codes on GA solving FVRP (benchmark + improved method)

2. Report for this project including explanation of the model, the methodology of the improvement, and numerical results, comparisons are needed.

Project Number: 10

Project Title: Future drought explorer system

Project Clients: Sanaa Hobeichi

Project specializations: Web Application Development;Big data Analytics and Visualization;Cloud Computing;Computer Science and Algorithms;Climate Data Science;

Number of groups: 2-3 groups

Main contact: Sanaa Hobeichi

Background:

Overview

The Future drought explorer system helps stakeholders (e.g., those working in agriculture, water management, climate policy) understand how drought conditions might change in Australia due to climate change. The system integrates future climate data from multiple sources, calculates or retrieves drought metrics from one or multiple drought indices, and displays whether drought conditions will increase, decrease, show no significant change, or remain unclear.

Drought conditions can be defined based on the number of drought events or the length of drought events; users are free to select the definition that matters most to them.

The primary goal is to provide a clear, accessible and interactive representation of potential changes in drought conditions in the future.

Requirements and Scope:

The project encompasses the development of a web-based system that allows users to explore projected drought changes across Australia.

Required Knowledge and skills:

The system must provide all core functionalities listed below and at least one novel functionality. The novel functionality can be one of those suggested here or a different one created by the students.

Core (required) functionalities

1. Interactive Map of Australia

Displays a clickable map subdivided into relevant regions (e.g., Natural Resource Management NRM regions).

2. Drought definition & Index Selection

Users can define drought change based on change in number of events or change in length of events

3. Drought index

- Users can select a drought index, such as:

Standardized Precipitation Index (SPI),

Standardized Precipitation Evapotranspiration Index (SPEI), or

Palmer Drought Severity Index (PDSI), or other available drought indices

- "Drought" or "non-drought" conditions are determined based on thresholds applied to the selected indices

4. Future timeframes

The clickable map displays how drought conditions may potentially change for a future time frame (e.g., near-term: 2020–2059, mid-century: 2040–2079, or late-century: 2060–2099) relative to a baseline period (e.g., 1980–2019).

5. Visualisation of the change

- The main display shows whether each region's drought conditions are increasing, decreasing, unchanged, or unclear compared to the chosen baseline period.

6. Multiple data source

The system integrates climate projection data from at least two providers, such as 2 climate simulations from CMIP5 or CMIP6 (Coupled Model Intercomparison Project).

7. On-click regional summary

Clicking on a region displays a summary table with at least four statistics. For example:

- Average number of drought months per decade in the baseline period.
- Average drought length in the baseline period.
- Projected change (increase, decrease, no change, or unclear) in the number of drought events.
- Projected change (percentage of increase, percentage of decrease) in drought length.

Additional (novel) functionalities

Below are suggestions for additional features that build upon the core idea of visualising how drought may potentially change:

8. Scenario selection

Users can compare drought change under different emission or climate scenarios (e.g., RCP4.5 vs. RCP8.5).

9. Threshold selection

Users can apply custom thresholds to classify conditions as drought or non-drought based on the selected indices.

10. Multiple future timeframes

Users can select and compare results across multiple future timeframes.

11. Agreement/disagreement between different drought metrics

The system provides a visual overlay highlighting areas where different metrics (change in drought length and change in the number of drought events) agree or disagree on the projected change (e.g., cross-hatching or coloured outlines).

12. Agreement/disagreement between different drought indices

The system provides a visual overlay highlighting areas where different drought indices agree or disagree on the projected direction of drought change.

13. Agreement/disagreement between different data sources

The system provides a visual overlay showing where different climate models (i.e. data sources) agree or disagree on drought change.

Expected outcomes/deliverables:

The sutdent are expected to produce a web-based visualisation system and provide a documentation.

Project Number: 11

Project Title: Web Interface for Hugging Motion Control of a Human-like Robot

Project Clients: Francisco Cruz Naranjo, Mahdi Bamdad

Project specializations: Software Development;Web Application Development;System/game Development;Computer Hardware and Networks;Robotics;Computer Vision;Human Computer Interaction (HCI);Mobile Application Development;

Number of groups: 2 group

Main contact: Francisco Cruz Naranjo, Mahdi Bamdad

Background:

Human-robot interaction is crucial for humanoids. Our Baxter robot operates through ROS1 (Robot Operating System) commands, precisely controlling its arms. Currently, students are working on interaction projects, and enabling Baxter to give hugs as an example of interaction has been discussed and implemented. Our next step is to enhance its perception by adding sensors, such as advanced vision capabilities, within a new user-friendly interface.

Requirements and Scope:

This project involves understanding the development tools for robot programming in a corporate environment. It includes assembling and programming a Baxter robot kit, developing a ROS-based web app, and applying advanced ROS concepts to control the robot motion within a containerized environment using continuous integration.

Required Knowledge and skills:

- Design and equip the robot with advanced sensors
- Software should allow real-time operation of robot's hardware including arms and sensors.
- It should support ROS for robot navigation, perception, and control.
- The software could be containerized (e.g. Docker) and development tasks should be automated (e.g. Jenkins).
- Check the integrity of the code (e.g. Continuous integration)
- Testing and validation to ensure the robot's performance meets safety and functionality.

Expected outcomes/deliverables:

The project results in a safe robot that gives you the perfect hug. Wrapping its dexterous robotic arms around you with none of the squeeze.

- Documentation: Detailed documentation of the software, including web interface.

- Containerized Environment and CI: Docker containers for the software and continuous integration pipeline.

* The project scope can be adjusted based on the group size. Don't worry about the large scope—we'll make sure everyone can contribute effectively.

Project Number: 12

Project Title: AI Driving Coach: Using Telemetry Data to Analyse Car Performance & Driver Emotion

Project Clients: Gelareh Mohammadi, Mahdi Bamdad

Project specializations: Software Development;Web Application Development;Mobile Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;System/game Development;Computer Hardware and Networks;Human Computer Interaction (HCI);Bioinformatics/Biomedical;

Number of groups: 2 group

Main contact: Gelareh Mohammadi, Mahdi Bamdad

Background:

We have designed and assembled a high-performance racing simulator, featuring a full cockpit setup, controllers, VR cameras, and a sensory device, all integrated with a fully licensed software package (SimHub). This newly established system is ready for both research and enjoyment.

Our project focuses on developing a driving coach/assistant that enhances racing performance by analyzing both car telemetry and driver emotions. A group member will serve as the test subject, making the experience both immersive and fun.

Project Goals: (1)Develop/Use a Custom SimHub Plugin – Enhance functionality by integrating external sensor data. (2) Forward UDP Telemetry for Processing –real-time driving data for analysis. (3) Integrate with an LLM (AI Race Engineer) – Enable hands-free coaching using voice commands and real-time insights. (4) Involving the emotion response and physiological measures (e.g. Heart-Rate)

Requirements and Scope:

Driving Coach Features: (1) Real-time Driving Assistance – Provides insights on car handling, race strategy, and performance. (2) Voice-Controlled Interaction – Communicate with the AI Race Engineer using natural language. (3) Hybrid Rule-Based & AI Engine – Combines a structured rule engine with GPT-powered natural language capabilities. (4) SimHub Integration – Bridges communication between the simulator and external software via UDP - API. (5) Dashboards – Displays performance metrics and driving recommendations.

Required Knowledge and skills:

(1) Software Development & AI Integration: Plugin Development , UDP & Network Programming

(2) Game Telemetry & Racing Simulation: SimHub API , Telemetry Data Analysis

(3) Sensors & Hardware Integration: Analyzing driving behavior, Speech Recognition , Text-to-Speech Synthesis

(4) Real-Time Data Processing/Visualization: Building live telemetry dashboards

(5) Testing and validation to ensure the performance meets functionality.

Expected outcomes/deliverables:

By combining sensor data, AI coaching, and real-time telemetry, this system provides an immersive and data-driven racing experience, helping drivers improve their skills while having fun! The project scope can be adjusted based on the group size. Don't worry about the large scope—we'll make sure everyone can contribute effectively on these items:

- Documentation: Detailed documentation of the software, including installation and usage instructions during testing and validation.

- Prototypes: Different design prototypes evaluated by the groups.

Project Number: 13

Project Title: Business Process for SMEs Utilizing E-invoicing APIs

Project Clients: Muhammad Raheel Raza

Project specializations: Software Development;Web Application Development;

Number of groups: 5 groups

Main contact: Muhammad Raheel Raza

Background:

o Streamline e-invoicing processes for SMEs by leveraging e-invoicing APIs.

o Test different assigned e-invoicing APIs and identify the endpoint functions on a GUI.

o Reduce manual intervention and errors associated with traditional invoicing methods.

o Enhance efficiency and accuracy in invoice generation, validation, and transmission

Requirements and Scope:

o Development of a business process flow for e-invoicing encompassing creation, validation and transmission categories.

o Design of a fully functional front-end application or GUI, testing the e-invoicing API via different endpoints and demonstrating the endpoint functions.

o Identification and demonstration of the API endpoint functions on GUI.

o Documentation of the implemented API service and API integration guidelines for SMEs.

Required Knowledge and skills:

o Selection of suitable e-invoicing APIs based on compatibility with SME systems, regulatory compliance, and cost-effectiveness.

o Test different selected/assigned e-invoicing APIs by writing clients and identify the endpoint functions on a GUI.

o The GUI should show each specific API endpoint to the user and produce results.

o Implementation of data encryption and privacy measures to safeguard sensitive invoicing information in the front-end.

Expected outcomes/deliverables:

- Fully functional source code.
- An in-depth report documenting the entire project process.
- A testing GUI interface showing each specific API endpoint functionality.
- API documentation provided by the selected e-invoicing service providers, including API endpoints, request/response formats, authentication methods & error handling procedures.
- Detailed documentation outlining the business process flow, API integration steps and usage guidelines for SMEs.

Project Number: 14

Project Title: AI Comment Moderation - RAG and Classification modelling

Project Clients: Nathaniel Lewis

Project specializations: Software Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Web Application Development;

Number of groups: 4 groups

Main contact: Nathaniel Lewis

Background:

To support review and moderation of student comment data, staff currently run a manual checking of thousands of student comments. AI and LLM tools are potentially well suited to moderation/checking of qualitative comment data and making recommendations for reviewers actions to take, reducing manual tasks and standardising approaches.

Requirements and Scope:

Develop a proof-of-concept (PoC) system that moderates student feedback comments using Retrieval-Augmented Generation (RAG) instead of custom training.

The system will:

-Ingest and store student feedback comments in a structured format.

-Retrieve past similar feedback examples from a vector database

-Use an LLM to classify new comments based on retrieved examples.

-Determine if human review is needed based on classification confidence.

-Provide an interface for manual review of flagged comments, and versioning, and saving of data.

-Provide recommendation for alternative approaches to supplement RAG approach with LLM pre-training

-Bonus points for Agent integration and LLM tool use for structured outputs

Key Constraints:

-No custom model training—only RAG-based techniques.

-Uses dummy data with a well-defined schema.

Required Knowledge and skills:

Data Ingestion, Storage, Management: The system will accept student feedback comments (dummy dataset) from 'Best Aspects' and 'Areas for Improvement' fields, storing them in a structured format for processing. IT should also allow versioning and export of pre and post datasets to file (csv, tsv, excel).

Schema Definition: A standardised input/output JSON schema will be used to ensure consistency across all components, making it easier to pass data between modules.

Vector Database for Retrieval: Feedback comments will be embedded using a pre-trained model and stored in a vector database (FAISS, Weaviate, or Pinecone) to enable similarity-based retrieval.

Retrieval Engine: When a new comment is received, the system will retrieve top-N most similar past comments to provide context for classification and moderation.

LLM-Based Classification: A pre-trained LLM (GPT-4, Claude, or Hugging Face model) will analyze the new comment in relation to retrieved examples and assign it a moderation category (e.g., Safe, Needs Review, Violent, Swearing, etc.).

Moderation Confidence Scoring: The system will generate a confidence score for each classification decision. If confidence is high, the comment will be automatically categorized; if confidence is low, it will be flagged for human review.

Human Review Interface: A simple web-based UI will allow moderators to view flagged comments, check retrieved examples, and confirm, override, or edit AI-generated classifications.

API Integration & Automation: The entire pipeline will be modular and API-driven, allowing seamless integration between data ingestion, retrieval, classification, and human review.

Expected outcomes/deliverables:

-Source Code (GitHub repo with documentation)

-Structured Dummy Dataset (JSON/CSV format)

-Vector Database with Sample Comments (FAISS/Weaviate setup)

-Basic API for Retrieval & Classification

-Simple UI for Moderation Review

-Project Documentation (setup guide, architecture)

-Final Report & Presentation (PoC demonstration)

Project Number: 15

Project Title: Social Mobile Application with in-app coins

Project Clients: reKro / Reuben Roy

Project specializations: Mobile Application Development;

Number of groups: 3 groups

Main contact: reKro / Reuben Roy

Background:

reKro is an early-stage startup with a team of 12.

Its currently part of an incubator program and has launched it v1 MVP, gearing up for its pre seed fund raise.

The team comprises of an operations team, a tech team (CTO, 3 devs, 2 UI/UX designers) and a Brand and Comms team.

Goals

To work with students on a product designed for students on campus.

To offer real life data and product development knowledge

To build on previous iterations

Requirements and Scope:

Students are to build a mobile application referencing and improving upon the current design

Required Knowledge and skills:

- Group/ personal chat application for Uni students

- Students can create and join chat groups

- Students can also attend Events in the app

- Students earn coins the more engaged they are within the application

- When you use partner brands/ allied services (on the app) , you can redeem these coins to get discounts on products and services

Use React Native for the Mobile application. Test it on different iOS, and Android devices.

Use Supabase or similar technology for database. Stripe API to handle payments.

Expected outcomes/deliverables:

Source code,

Documentation,

Working application prototype

Project Number: 16

Project Title: AI Chatbot that answers students inquiries

Project Clients: reKro/Liam Baga

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 3 groups

Main contact: reKro/Liam Baga

Background:

reKro is an AI powered community builder tool that connects students with each other and their allied services.

It comprises a team of 12

An operations team, a tech team(CTO,3 devs, 2 UI UX designers), and a brand and comms team.

Goals

Students are to work with the provided resources and data to build an AI chatbot

Requirements and Scope:

- Utilise publicly available University data for the AI chatbot's custom knowledge base.

- The chatbot should answer a student's inquiry to the best of it's knowledge

- The quality of your data would determine chatbot's reply quality and accuracy

Required Knowledge and skills:

- Use Retrieval-Augmented Generation to create external data, retrieve relevant information, augment the LLM prompt, and Update external data

- Utilise the RAG process to power the AI Chatbot that chats with uni students. The chatbot should respond about uni, or course specific questions

Expected outcomes/deliverables:

Source code,

Documentation,

Working application prototype

Project Number: 17

Project Title: Statsplainer: An online reader application to understand statistics in academic literature

Project Clients: Main: Benjamin Tag, Co: Simo Hosio

Project specializations: Human Computer Interaction (HCI);Artificial Intelligence (Machine/Deep Learning, NLP);Web Application Development;

Number of groups: 3 groups

Main contact: Main: Benjamin Tag, Co: Simo Hosio

Background:

Academic papers, industry reports, and technical documents often contain complex statistical data that can be difficult to interpret for non-experts. Understanding statistical concepts is crucial in fields such as healthcare, business, policy-making, and research, but many readers struggle to comprehend the meaning and implications of statistical results.

Existing solutions like statistical glossaries and educational videos provide static explanations, but they lack interactivity and contextual relevance. Recent research in Human-Computer Interaction (HCI) and AI has explored ways to simplify complex data through interactive explanations.

Project Goal:

The goal of StatSimplify is to build an interactive online PDF reader that allows users to:

✅ Upload a PDF document

✅ Highlight a paragraph containing statistical information

✅ Automatically generate intuitive explanations in a sidebar, including:

- A simplified breakdown of the statistical terms
- A real-world analogy for better understanding
- An "Explain Like I'm 5" (ELI5) explanation of its significance and application

This project will make statistical content more accessible, interpretable, and user-friendly, particularly for students, researchers, and professionals who may not have a strong background in statistics.

Requirements and Scope:

The project will focus on the following core components:

In-Scope:

- PDF Upload & Processing – Users can upload PDF documents and extract text

- Statistical Text Recognition – Identify statistical terms, numbers, and concepts within a highlighted paragraph

- Automatic Explanation Generation – Provide explanations, real-world analogies, and ELI5 descriptions

- Interactive UI with Sidebar Display – Users can highlight a paragraph, triggering an interactive sidebar with explanations

- Expandable Knowledge Base – The system should improve over time by incorporating additional statistical explanations

- Web-Based Deployment – The tool will be accessible via a web application

Additional, if time allows:

- Full AI-powered explanation generation (Instead, we will use predefined templates and rule-based methods)

Required Knowledge and skills:

Detailed Features & Specifications:

1. PDF Processing & Text Extraction

- Users can upload PDFs
- Extract and display text content
- Highlighting functionality for users to select specific paragraphs

2. Statistical Concept Recognition

- Identify statistical terms and values (e.g., p-values, confidence intervals, regression coefficients)
- Extract and categorize statistical terms from text using Natural Language Processing (NLP)
- Use open-source repositories like Statslator.js and Charagraph

3. Explanation Generation Engine

- Provide three levels of explanation:

1. Basic definition of the statistical concept

2. Real-world analogy for better understanding

3. “Explain Like I’m 5” (ELI5) explanation

- Develop a predefined knowledge base of statistical concepts and explanations
- (Optional) Experiment with AI models to enhance explanation quality

4. Interactive UI with Sidebar Display

- Users can highlight a paragraph
- Sidebar dynamically provides:
- Recognized statistical terms
- Corresponding explanations and analogies
- UI should be intuitive and minimalistic for easy interaction

5. Web-Based Application

- Develop a responsive web app using React.js or Vue.js
- Backend API to handle text processing and explanations
- Deploy using AWS, GCP, or Firebase

Expected outcomes/deliverables:

=> A functional web-based prototype, including:

- PDF Upload & Highlighting Feature
- Explanation Sidebar with statistical concept breakdown
- Interactive UI for easy user engagement

=> Codebase (open-source) hosted on GitHub

=> Technical Documentation

- Architecture and implementation details
- API documentation

=> User Guide & Demo Video

- Instructions on how to use the tool

Project Number: 18

Project Title: AI chatbot for future energy workforce development

Project Clients: Dr. Hua Chai

Project specializations: Software Development;Web Application Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;Human Computer Interaction (HCI);

Number of groups: 5 groups

Main contact: Dr. Hua Chai

Background:

Achieving net-zero goals requires expanding the clean energy workforce. However, a major challenge is the lack of accessible and accurate information about pathways into clean energy roles. This project aims to develop an AI-driven chatbot that supports a broad spectrum of users (e.g., K–12 students, prospective and current university students, sector shifters, returning or retired workforce, to experienced professionals), who want to learn about and engage in clean energy careers. The chatbot will offer tailored recommendations on education programs (including degree pathways at UNSW), career options, and upskilling/reskilling opportunities. For instance, younger audiences might access simplified, real-world STEM applications, while current or prospective university students can explore relevant degree programs, and experienced professionals can receive advanced career guidance.

Requirements and Scope:

• Develop an AI chatbot capable of understanding diverse user needs and backgrounds, with structured conversation flows that adapt to each user's context.

• Provide personalized career and education guidance across different life stages by offering tailored STEM insights, academic recommendations, reskilling pathways, and upskilling opportunities for individuals transitioning into or advancing within the clean energy sector.

• Implement or prototype automated real-time data scraping to keep job postings and industry demand insights up to date (e.g., job boards, industry sources).

• Create an intuitive, web-based chatbot interface to be delivered within the project timeline.

• Lay the foundation for continuous improvement through usage analytics and easy expansion to new user groups or institutions.

Required Knowledge and skills:

• Implement or integrate NLP frameworks (e.g., Rasa, GPT-based models) to handle varied user queries with context-awareness.

• Structured Database: Develop a robust data model for university programs, career pathways, job postings, and user profiles.

• Automated Data Gathering: Explore web scraping or API-based solutions to fetch live job postings, ensuring timely and relevant career guidance.

• Front-End: Build a responsive UI/UX that accommodates multiple user demographics.

• Back-End: Establish secure, scalable, and well-documented server-side functionalities and APIs.

• Ensure the system can handle concurrent requests while safeguarding user data and maintaining system integrity.

• Provide accurate, unbiased responses and uphold strong data privacy practices.

Expected outcomes/deliverables:

• A functional AI Chatbot (Web-Based) that offers tailored interactions for identified user segments.

• A well-structured, version-controlled codebase (GitHub, GitLab, etc.)

• Documentation including technical documentation and user & maintenance guides

• Project Report & Presentation: An overview of the design process, technical approaches, outcomes, and potential for future improvements.

Project Number: 19

Project Title: Automated E8 Compliance and Remediation

Project Clients: Jesse Laeuchli, Arash Shaghaghi

Project specializations: Software Development;Web Application Development;Big data Analytics and Visualization;Cloud Computing;Computer Hardware and Networks;Security/Cyber Security;

Number of groups: 3 groups

Main contact: Jesse Laeuchli, Arash Shaghaghi

Background:

Compliance with security standards is an important goal in cyber security. We are interested in developing software that can be used to automate compliance with ASD's essential 8.

Requirements and Scope:

The project would aim to develop a set of tools which would

1). Collect data about the state of the clients network and hosts

2). Analyse the collected data in order to determine compliance level, based on the ASD checklist.

3). Provide the results in the form of a dashboard. We have a dashboard which reports compliance of other standards, so integration with this dashboard would be ideal.

4).Where appropriate the tool would seek to remediate lack of compliance.

Required Knowledge and skills:

The project has three main parts

1). Development of a data collection application which can be deployed to clients and servers, which would read different system options and configurations, as well as different network options and configurations. This application would collect the data required to determine if E8 is being met at a given maturity level. This requires reading many configuration settings at the system and network level, and would need to make extensive use of system APIs and Powershell to obtain this information.

2). A server application which will manage the deployed data collection clients, and then use the collected data to determine if the network meets a given control at a given maturity level. This application could be written in anything, but will require some understanding of networking as well as general software development skills.

3). A web dashboard, where the results can be displayed. This dashboard could be stand alone, but ideally would integrate with our existing dashboard. This will require an understanding of web application development.

Expected outcomes/deliverables:

Source code, and a deployment guide.

Project Number: 20

Project Title: Development of AI Chatbots for Student Wellbeing

Project Clients: Jihyun Lee (Professor in the School of Education, UNSW)

Project specializations: Software Development;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 3 groups

Main contact: Jihyun Lee (Professor in the School of Education, UNSW)

Background:

This project aims to develop AI (Artificial Intelligence)-based chatbots to support student wellbeing.

In this project, students can develop AI chatbots aiming to help students in terms of self-awareness, self-management, social awareness, relationship skills, and responsible decision-making (see this website: https://casel.org/fundamentals-of-sel/what-is-the-casel-framework/) or student learning engagement (motivational, emotional, behavioral), learning strategies (cognitive, meta-cognitive, self-management), and social influences from teachers, peers, and family networks (see Lee & Shute, 2010). Students can choose one of these aspects of student wellbeing or combination of them.

Recent research demonstrates the potential of AI-driven chatbots (Hsu et al., 2022; Xia et al., 2023) to enhance student learning and address mental health concerns (He et al., 2023; Li et al., 2022). However, most studies on AI's impact on wellbeing have focused on clinical populations, leaving a significant gap in understanding its effects on the wellbeing of non-clinical, student populations. Student wellbeing requires a different focus than wellbeing of clinical patients. Wellbeing is a multidimensional construct, involving social-emotional learning (CASEL, 2020), encompassing student engagement (motivational, emotional, behavioral), learning strategies (cognitive, meta-cognitive, self-management), and self- and social-awareness of influences from teachers, peers, and family networks (Lee & Shute, 2010).

Requirements and Scope:

Students will develop AI chatbots (software tools), targeted for secondary school students or university students. The project scope is the design and development of software/system application.

Required Knowledge and skills:

AI chatbots (software tools) to be developed in this project should be user-friendly, easy to use with well-structured design. AI chatbots should allow users to have conversations, guidance, and support in the area of learning, social-emotional learning, and personal growth.

Expected outcomes/deliverables:

Source code, documentation, and user-friendly AI systems for student wellbeing

Project Number: 21

Project Title: Code-Enhanced Vulnerability Intelligence Alignment

Project Clients: Jiaojiao Jiang

Project specializations: Artificial Intelligence (Machine/Deep Learning, NLP);Security/Cyber Security;Computer Science and Algorithms;

Number of groups: 2 group

Main contact: Jiaojiao Jiang

Background:

Vulnerability intelligence is crucial for cybersecurity, but current databases struggle with fragmented and inconsistent information. This project proposes enhancing vulnerability alignment by combining code-level analysis with traditional text-based reports. By integrating semantic data from code commits, patches, and snippets with vulnerability metadata (like affected products, attack vectors, and impacts), the framework aims to enable more accurate vulnerability intelligence consolidation and improved risk assessment.

Requirements and Scope:

This project will build a system that combines code analysis with vulnerability reports. The system will extract key vulnerability aspects from text-based reports using state-of-the-art language models and will simultaneously analyze related vulnerable code snippets (PoC) using a code-aware model. These dual data streams will be constructed into a multi-relational knowledge graph. A graph neural network-based alignment module will then be used to accurately align vulnerability entries across heterogeneous sources to support dynamic risk assessments and tailored patch prioritisation recommendations.

Required Knowledge and skills:

Extract code semantics to analyze vulnerability-related commits, patches, and code snippets.

Build a multi-relational knowledge graph by combining text-based vulnerability data with code semantic embeddings.

Implement a GNN-based alignment system for accurate vulnerability matching and dynamic risk assessment.

Expected outcomes/deliverables:

A working prototype system that integrates code-level semantic analysis with text-based vulnerability intelligence for robust alignment.

Source Code thorough documentation and a user guide.

Project Number: 22

Project Title: Automatic segmentation of computed tomography scans using MONAI and 3DSlicer

Project Clients: Luca Modenese

Project specializations: Software Development;Artificial Intelligence (Machine/Deep Learning, NLP);Bioinformatics/Biomedical;

Number of groups: 2 group

Main contact: Luca Modenese

Background:

Orthopaedic applications require segmentation of radiological scans, usually from computed tomography, to extract three-dimensional bone shapes that can be used for pre-surgical plans.

The goal of this project is to use publicly available datasets of segmented medical images (several of which listed at https: //github.com/modenaxe/awesome-biomechanics) to train a deep-learning model for automatically segmenting CT scans of the lower limb in the MONAI framework. The trained model will be made available to non-technical users through 3DSlicer so they can segment their own images and further refine the model through MONAI label.

MONAI (https: //monai.io/) is the open-source framework that is becoming the de facto stardard for developing artificial intelligence solutions related to medical images.

3DSlicer (https: //www.slicer.org) is a widely used open-source application in which it is possible to prototype solutions involving processing of medical images.

Requirements and Scope:

The project is limited to:

- the training of the MONAI model,

- the creation of the Slicer extension (using the extension wizard or based on existing similar plugins)

- the working setup of a MONAI label server.

The datasets will be provided but the data might need format transformations.

Required Knowledge and skills:

- The segmentation model will be implemented in the MONAI framework, ideally using the autoseg3d.

- the Slicer extension to be similar to this one: https://github.com/lassoan/SlicerMONAIAuto3DSeg

- I would like the trained model to be made available to users also through MONAI Label for continuous refinement when further data become available.

Expected outcomes/deliverables:

- a trained nnUnet or auto3dseg model to use in 3D Slicer (weights)

- a Slicer plugin working with the provided model

- a MONAI label server setup to use the trained model and extend it

Project Number: 23

Project Title: Conference program committee recommender system

Project Clients: Serge Gaspers

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;

Number of groups: 3 groups

Main contact: Serge Gaspers

Background:

The client is a program committee chair and needs to create a program committee for a computer science conference.

A program committee (PC) consists of members of the academic community and is in charge of deciding which submitted papers get accepted into the program of the conference.

This project develops a recommender system in a real-world scenario and will consist of a software system to help create such a PC.

Requirements and Scope:

This includes

- identifying topics of expertise that the PC needs to cover, based on textual data from webpages or calls for papers from previous editions of the conference

- predicting how many papers will be submitted in each topic based on the publication data of previous years (match papers to topics based on title, abstract, keywords)

- recommending academics to the PC

- recommending additions to a partially formed PC

- facilitating invitation via email to an up-to-date email address

Data will be obtained from DBLP, Google Scholar, ORCID, academic web pages, conference web pages, conference calls for papers.

The system should work for any computer science conference.

Required Knowledge and skills:

It is expected that this system will be used in multiple rounds to process the information of who accepst/declines the invitation to be on the PC.

The system should allow to recommend academics to the PC

- covering the needed expertise, by taking into account that each submitted paper will be assigned to multiple (typically 3) PC members

- prioritise academics who have published often at the conference (especially recently); this data should be obtained from [DBLP](https: //dblp.org/)

- prioritise diversity in terms of seniority, location, gender (this data needs to be scraped/learned/estimated)

- handle manual inclusion/exclusion from the PC (invited speakers are typically excluded, academics who have already declined/accepted the invitation to the PC)

- handle manually entered estimates of reliability of academics

It should visualize these recommendations in a succinct way and facilitate the selection of the next group of academics to be invited to the PC.

Expected outcomes/deliverables:

Functioning system, open source code, documentation, user guide.

Project Number: 24

Project Title: Online Audio Quality Rating System

Project Clients: Helen Paik, Lina Yao, Haowei Lou

Project specializations: Software Development;Web Application Development;Mobile Application Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;

Number of groups: 3 groups

Main contact: Helen Paik, Lina Yao, Haowei Lou

Background:

Unlike deep learning tasks such as classification and image generation, audio-related research—particularly in speech and music—relies heavily on human subjective assessment. Listeners evaluate audio clips based on predefined metrics like naturalness, intelligibility, and quality, but current methods face limitations: limited listener pools, subjective variability, and inconsistent assessments, impacting research reliability.

To address these challenges, we propose an online speech rating system that connects researchers with diverse, skill-matched listeners. This scalable platform enables researchers to upload audio samples for evaluation using standardized metrics, reducing bias and improving efficiency in subjective speech assessment.

Requirements and Scope:

The project aims to develop an online speech rating system that enables researchers to upload audio samples and obtain structured, large-scale evaluations from skill-matched listeners. The system seeks to reduce subjective biases, improve evaluation efficiency, and ensure more reliable and diverse assessments compared to traditional methods.

Required Knowledge and skills:

1. Account Management

- Users can register and manage their accounts.
- Researchers gain access to project management and data analysis tools.
- Listeners can create and update their profiles for participation in evaluations.

2. Project Management

- Researchers can create and manage projects.

- Ability to upload audio clips in various formats.
- Assign tags to audio files for evaluation purposes (tags remain hidden from listeners but are used for result aggregation).
- Define evaluation metrics such as naturalness, intelligibility, and clarity for listeners to assess.

3. Listener Profile Management

- Listeners can maintain a detailed profile, including:
    - Background information (e.g., academic or professional expertise)
    - Native language and additional language proficiency
    - Experience in speech evaluation
    - Demographics (age, gender, country of residence, etc.)
- Profiles help optimize listener selection for fair and diverse evaluations.

4. Audio Rating System

- User-friendly rating interface with:
    - Mean Opinion Score (MOS) scale (1-5)
    - Custom researcher-defined evaluation metrics
    - Optional textual feedback submission
- Built-in validation mechanisms to ensure high-quality and unbiased ratings.

5. Reward System

- Encourages listener engagement through a point-based reward system.
- Listeners earn points based on:
    - Participation (number of valid evaluations completed)
    - Consistency (alignment with expert assessments)
    - Accuracy (ratings matching expected trends)
- Rewards can be redeemed for gift cards, discount codes, premium research access, or monetary incentives.

6. Smart Audio Allocation System

- An AI-powered automatic allocation system assigns audio clips to the most suitable listeners.
- Ensures fair and balanced evaluations by:
  - Matching audio to listeners based on profile data (e.g., language proficiency, demographics).
  - Preventing subjective bias by distributing clips across diverse listener groups.
- Aggregates and presents listener demographics after evaluation completion:
  - Example: "This project was rated by 100 listeners from 10 countries: 90 native speakers, 10 fluent non-native speakers, 50 male, 50 female."

7. Data Analysis & Reporting

- Comprehensive analytics dashboard for researchers.
- Aggregated rating data visualization for easy interpretation.
- Exportable reports in multiple formats: CSV, JSON, PDF.
- Statistical insights such as:
  - Mean, standard deviation, and confidence intervals of ratings.
  - Rating distribution across different listener demographics.
  - Comparative analysis across evaluation criteria.

8. Anomaly Detection System

- Detects and removes outliers and unreliable ratings to maintain data integrity.

Expected outcomes/deliverables:

source code, hosting, user guide

Project Number: 25

Project Title: Developing software application for decomposing and editing high-density EMG experimental data into motor unit firing patterns

Project Clients: Luca Modenese / Andrea Sgarzi

Project specializations: Software Development;Bioinformatics/Biomedical;Computer Science and Algorithms;

Number of groups: 2 group

Main contact: Luca Modenese / Andrea Sgarzi

Background:

High-density electromyography (HDEMG) is an experimental techniques used to non-invasively record the electric potential generated by contracting muscles. The experimental signals, once processed through a step of decomposition and usually some manual editing/checks, provide information about individual motor unit and their signal to the muscle.

There are currently no high-quality (and ideally open-source) software tool to process these measurements, but only MATLAB toolboxes like DEMUSE (https: //demuse.feri.um.si) or MUEdit (https://github.com/simonavrillon/MUedit). The former option is closed-source and commercial.

Requirements and Scope:

The project scope consists in developing a software with equivalent functionality to DEMUSE and MUEdit for decomposing, editing and exporting the motor unit firing patterns detected from HDEMG measurements. All the required operations to implement are available on public github repository, so there will be no research development to perform, but just porting from MATLAB, connecting from existing Python modules and graphical user interface development.

Required Knowledge and skills:

- The software should consist of a graphical user interface similar to MUEdit or DEMUSE (see links in background).

- the software should allow the choice between the decomposition algorithm implemented in MUEdit (https: //github.com/ciaragibbs/MUEdit_Python) and more recent approaches like SCD (https://github.com/AgneGris/swarm-contrastive-decomposition)

Expected outcomes/deliverables:

Outcome of the project is a GUI from which the user can:

* visualise/inspect the collected experimental data

* decompose the HDEMG in motor unit spike trains choosing the algorithm between those currently available in public repositories

* manually edit the spike trains if required

* saving the results of the processing in a format convenient for further analysis in Python

Project Number: 26

Project Title: Web Platform for Microservices and APIs

Project Clients: Dr. Basem Suleiman

Project specializations: Web Application Development;Software Development;

Number of groups: 2 group

Main contact: Dr. Basem Suleiman

Background:

The goal of this project is to design and develop a web application for creating and maintaining microservices and APIs.

Requirements and Scope:

Users accounts and profile management

Scraping data from websites and other sources

Data storage and retrieval using different formats

System management

Data collection by end users

Required Knowledge and skills:

Users should be able to create their accounts and log using email, and social media. Users must be verified

Implement user roles including Admin, service providers, and service consumer, and public user with appropriate access controls.

Administrator user should have the full privilege in the system and should be able to manage access control and oversee the platform.

Users should be able to manage their profiles, including personal details, subscription preferences, and API usage data.

Service providers can add their APIs/microservices with structured documentation of technical and non-technical details. They can add tags to their services as well.

Service providers, and admin users, can automatic extraction of service/API details from a service description and source code (e.g., using Swagger or other documentation)

Service providers can manage multiple versions of their services/APIs.

Service providers can respond to ratings/reviews on their services/APIs

Users can search for microservices/APIs based on keywords, category, provider, etc

Recommend related/similar services in search results

Service providers can see dashboard – summary analytics of key insights of their services

Users can filter microservices/APIs based on type, category, date, version and other important information.

Service consumers, and public users, can search, browse the details of, and subscribe to services/APIs (basic and advanced search with filters)

Service consumers can test the service with input parameters

Service consumers can subscribe to services/APIs, so they receive notifications for any updates

Service providers can track services/APIs analytics for each service (e.g., subscriptions, rating, search).

Expected outcomes/deliverables:

- Robust and scalable architecture supporting seamless user experience in web platforms and other devices.
- Fully functional application (robust, secure, and performance)
- Project documentation including source code, system architecture and design, UI/UX, different types of testing, user manuals and guide, technical guide.

Project Number: 27

Project Title: Web Application for Project Contests

Project Clients: Dr. Basem Suleiman

Project specializations: Software Development;Web Application Development;Cloud Computing;

Number of groups: 2 group

Main contact: Dr. Basem Suleiman

Background:

Project contests can be useful for creating environment for creativity and innovation. Managing such contests can be challenging due to the need for efficient handling of project submission, evaluation, and communication. This project aims to develop a web application that supports the management of project contests, to organizers to run successful events and for participants to engage effectively.

Requirements and Scope:

The web application will provide a platform for managing project contests which include the following:

- User and profile management
- Contest creation and management
- Project submission and tracking
- Judging and evaluation of projects
- Communication for organizers and participants
- Reporting and analytics

Required Knowledge and skills:

Users

- Administrator user should have the full privilege in the system and should be able to manage and oversee the platform.
- Users include contest organizers, participants, and public users.
- Users with different roles and responsibilities can create their account, login and operate the system based on their roles and responsibilities.

- Users can edit/maintain their profiles.

Organizers

- Create/register their account (and verify it) and login.
- Create and maintain organizer profile.
- Create and maintain a new project contest and specify the details of the contest
- Search and view project contests (own and others) and its details, and filter them
- Search and filter contest's participants and their project submission.
- Assess project submissions through specific procedure
- Rate and recognize the participants.

Participants / public users

- Create/register account and login.
- Create and maintain profile.
- Search and browse project contests, and submitted projects to a contest, and refine the results with search filters.
- View full details of any project contests, and project details submitted to a contest.
- Submit a project for a contest, and maintain their project submission
- Notifications of contests and projects of interest.
- View rating and feedback on own projects.
- Review and rate experience once completing a contest.
- Receive and view recognition of contests completion/participation.
- Rate and vote for projects

Expected outcomes/deliverables:

Expected Outcomes:

- Robust and scalable architecture supporting seamless user experience across web and mobile platform.
- Fully functional application (robust, secure, and performance)

- Project documentation including source code, system architecture and design, UI/UX, different types of testing, user manuals and guide, technical guide.

Project Number: 28

Project Title: Stronger Brains: Cross-Platform App for Brain Training & Education

Project Clients: Gelareh Mohammadi, Wenjie Zhang, Wendy Haigh

Project specializations: Software Development;Mobile Application Development;Cloud Computing;Security/Cyber Security;Web Application Development;Computer Science and Algorithms;Human Computer Interaction (HCI);

Number of groups: 2 group

Main contact: Gelareh Mohammadi, Wenjie Zhang, Wendy Haigh

Background:

Stronger Brains is an Australian not-for-profit organization dedicated to breaking the cycle of intergenerational disadvantage through brain-based education and training programs. Research in neuroscience, led by Dr. Michael Merzenich, has shown that the brain can be trained and strengthened, particularly for youth affected by trauma and toxic stress.

Currently, Stronger Brains has a web-based program that provides brain training and educational support for disadvantaged youth. To expand accessibility and impact, the organization aims to develop a cross-platform mobile application that integrates with its existing online portal and Brain Training App.

Requirements and Scope:

Below is the full scope of the project, but given the short duration of the project, this scope could be adjusted:

- Develop a Cross-Platform Mobile App

- The app should be available on iOS and Android.

Enable Offline Access

- The workbook content (100 lessons across 20 weeks) should be accessible without an internet connection.

Secure Online Brain Training

- Internet-based brain training exercises should be available, but with a firewall restricting general internet access.

Support Student & Teacher Functionality

- Students can engage with lessons and brain training activities.

- Teachers/Administrators can track student progress, including hours spent, exercises completed, and responses to daily questions.

- Seamlessly Integrate with Existing Backend

- The new app must connect to the current web portal and database.

Required Knowledge and skills:

1. Platform & Compatibility

- Cross-Platform Support: The app will be developed for both iOS and Android using a cross-platform framework (e.g., Flutter, React Native) replicating the current web application of the stronger brains and should be functional.

- User Interface (UI) & User Experience (UX): A simple, intuitive, and engaging design optimized for students and teachers.

2. User Roles & Access Levels

A) Student Users (Primary Users)

- Access Training Modules: Students should be able to access and complete brain training exercises.

- View Progress: Students can track their own learning progress.

B) Teachers/Administrators (Secondary Users)

- Dashboard & Analytics: Admin users can view student engagement, brain training hours, and exercise completion.

- Monitor Student Progress: Teachers can track individual and group progress, including responses to daily questions.

C) System Administrators (Stronger Brains Team)

- Manage User Access: Ability to assign and modify student/teacher accounts.

- Data Management: Maintain backend data, user progress, and system updates.

3. Offline Functionality

- Workbook Lessons Accessible Offline

4. Online Functionality & Security

- Firewall for Brain Training Component; The app must allow students to access internet-based brain training activities while restricting access to any other parts of the internet.

5. Integration with Existing System

- Data Syncing: All user progress, exercises, and reports should be stored in the existing backend/database.

- API Integration: The app should use APIs to fetch and sync data with the current system.

Expected outcomes/deliverables:

Fully functional Stronger Brains Cross-Platform App

Source Code & Documentation

User Manual for students, teachers, and administrators

Final Report & Presentation to showcase project outcomes

Project Number: 29

Project Title: English to Indian Sign Language Translator

Project Clients: Client: Suvrat Chaturvedi, Supervisor: Sonit Singh, Co-Supervisor: Sanjay Jha

Project specializations: Software Development;Computer Vision;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 2 group

Main contact: Client: Suvrat Chaturvedi, Supervisor: Sonit Singh, Co-Supervisor: Sanjay Jha

Background:

Researchers working in the domain of British Sign Language, American Sign Language etc have achieved great success. The same is lacking for the Indian Sign Language. The goal of this project is to assess whether the sign detection research done for BSL or ASL can be readily applied for ISL or if not then where are the current boundaries and how these can be further pushed out.

Requirements and Scope:

The scope of this project is limited to translation of English text into equivalent ISL gesture video using a skeleton avatar. In order to achieve this goal, it would use the ISL-600 dataset created by Longtail AI Foundation. The scope is further limited by subsetting the English words that the model would process and not be a general purpose translator from English to ISL.

We'll identify a few English sentence subset and then the target ISL videos that play out the corresponding Signs. This would serve as the gold standard for the project. In our project, we'd implement our model limited to the test cases defined above and then outcome would be verified against the gold standard ISL video.

Required Knowledge and skills:

In this project, we'd be required to use NLP for English language processing, followed by mapping of English word structure into ISL signs. Once done, the sign keypoints will be applied on the Avatar skeleton to play out the signs. This implementation will require development in Python, will require Compute for ML tasks.

Expected outcomes/deliverables:

We would deliver the following:

1. Source code.
2. Output videos
3. Comparison matrix of our outcome with the gold videos to demonstrate accuracy.
4. Direction for the future work.

Project Number: 30

Project Title: SDG Knowledge System: Web & Mobile Development Project

Project Clients: Associate Professor Veronica Zixi Jiang, School of Marketing, UNSW Business School. Chengbin Feng, Alumnus of UNSW Computer Science undergraduate, Academic Tutor in Information System.

Project specializations: Software Development;Web Application Development;Mobile Application Development;Big data Analytics and Visualization;Human Computer Interaction (HCI);

Number of groups: 3 groups

Main contact: Associate Professor Veronica Zixi Jiang, School of Marketing, UNSW Business School. Chengbin Feng, Alumnus of UNSW Computer Science undergraduate, Academic Tutor in Information System.

Background:

We developed the SDG Knowledge System (https: //sdg.unswzoo.com/) to provide comprehensive and relevant information on the United Nations' 17 Sustainable Development Goals (SDGs).

The SDG Knowledge System currently includes:

- SDG Education Database: With over 3,000 entries, this extensive collection offers in-depth insights into SDG-related education.

- SDG Action Database: Featuring 1,500+ curated items, this database highlights actionable SDG plans and real-world examples, supporting the effective implementation of sustainable practices.

- SDG Keyword Search: Users can search specific words or phrases to determine their relevance to the 17 SDGs and 169 targets.

- SDG Expert Chatbot: Designed to assist users in quickly finding relevant SDG information and brainstorming impactful SDG action plans.

This system empowers individuals, educators, and organizations to engage meaningfully with the SDGs, fostering informed decision-making and real-world action.

Requirements and Scope:

It is ok that a student group can only complete two or three of the project tasks.

Task 1 (most important): Implement a User Management System – Develop a secure and efficient user management system to enable personalized experiences and data tracking.

Task 2 (most important): Enable SDG Action Plan Creation – Allow logged-in users to create SDG action plans using an "Impact Design Analysis" form, providing a structured approach to planning and assessing sustainable initiatives.

Task 3: Develop a Mobile App – Create an accessible and user-friendly mobile application to enhance the reach and usability of the SDG Knowledge System.

Task 4: Enhance Website Accessibility & Design – Improve the website's accessibility, aesthetics, and user experience to ensure seamless navigation and engagement.

Required Knowledge and skills:

1. Implement a User Management System

• User Registration & Authentication

o Users can register via email, mobile number, Google, Apple, or Facebook accounts.

o Implement robust password policies for security.

• Profile & Group Management

o Users can manage their profiles, including name, contact information, job title, and department.

o Implement grouping functionalities, allowing teams to collaborate on SDG action plans.

• Data Privacy & Compliance

o Ensure adherence to relevant data protection regulations (e.g., GDPR, CCPA).

o Implement secure data storage and encryption to protect user information.

• Activity & Search History Tracking

o Save user activities, browser histories, search histories, and form inputs to enhance personalized experiences and analytics.

2. SDG Action Plan Creation & Management

• Impact Design Analysis Form

o Logged-in users can create, save, and edit SDG action plans using a structured form.

• Draft & Revisit

o Users can leave the form and return later to edit their responses.

• Slide Generation & Downloads

o Users can generate slides of their SDG action plans using predefined templates.

o The generated slides can be downloaded for presentations or reports.

3. Develop an Accessible Mobile App

• Create a mobile application that mirrors the website's functionality while ensuring seamless access and usability on smartphones and tablets.

• Ensure performance optimization and responsiveness across different devices and operating systems.

4. Enhance Mobile app and Website Accessibility & User Experience

• Improve accessibility by implementing:

o Good color contrast for readability.

o Adjustable text sizes to accommodate different visual needs.

o Clear and intuitive navigation to enhance usability.

o Support for assistive technologies (e.g., screen readers).

o Appropriate tap target sizes for better mobile interactions.

o Consistent layouts for an intuitive experience.

o Alternative input methods for users with disabilities.

o Compliance with WCAG (Web Content Accessibility Guidelines) to ensure full accessibility.

This project will create a robust, accessible, and user-friendly SDG Knowledge System that enhances engagement, collaboration, and action toward achieving the United Nations' 17 Sustainable Development Goals.

Expected outcomes/deliverables:

Source Code: All the front-end and back-end code developed for the website, ensuring it is well-organized, reusable, and efficient. This will include HTML, CSS, JavaScript, and any server-side programming languages or frameworks used.

Documentation: Comprehensive documentation outlining the structure of the website, key features, and functionalities. This will also include a breakdown of the website's development process, detailing the tools, technologies, and methodologies used. Code documentation will also be provided for easy future updates or modifications.

User Guide: A clear, concise guide that explains how to navigate and use the website's key features. This will be aimed at users looking to understand the SDG content, interact with tools, or contribute information. It will also include troubleshooting tips and frequently asked questions.

Final Report: A summary of the project, outlining its objectives, development process, challenges faced, and how the final product contributes to advancing SDGs through user engagement.

Testing Results: Reports on website performance, usability testing, and feedback gathered from users to ensure the site meets accessibility and functionality standards.

Project Number: 31

Project Title: Ontology-Enhanced PCA Models Using Eurofidai ESG Dataset

Project Clients: Mingqin Yu

Project specializations: Software Development;Computer Science and Algorithms;Big data Analytics and Visualization;

Number of groups: 3 groups

Main contact: Mingqin Yu

Background:

The integration of ontology-driven methodologies with Principal Component Analysis (PCA) can significantly enhance ESG metric interpretation and risk assessment. The Eurofidai ESG dataset, containing 105 raw risk data metrics across global companies, provides an ideal dataset for this study. The primary objective of this project is to develop an ontology-enhanced PCA model that improves data structuring, dimensionality reduction, and interpretability in ESG analysis.

This project aligns with the ontology-driven architecture proposed by Yu, Rabhi, & Bandara (2024), which emphasizes semantic standardization and structured ESG knowledge representation. Through this study, students will gain hands-on experience with machine learning, ontology engineering, and sustainability analytics.

Requirements and Scope:

2.1 Scope

Phase 1: ESG Data Preprocessing and Ontology Integration

This phase focuses on structuring the Eurofidai ESG dataset using ontology-based ESG categorization and preparing the data for PCA analysis.

Task 1: Data Exploration

1) Load and explore the Eurofidai ESG dataset, identifying key features, missing values, and anomalies.

2) Conduct summary statistics and visual analysis to understand metric distributions and data variability.

Task 2: Data Preprocessing and Normalization

1) Handle missing values using data imputation techniques.

2) Standardize numerical ESG metrics to ensure comparability across dimensions.

3) Convert categorical ESG features into ontology-compatible representations.

Task 3: Ontology Integration

1) Utilize ESG ontology models from Yu et al. (2024) to categorize ESG metrics into Environmental, Social, and Governance (E, S, G) groups.

2) Use SPARQL queries to retrieve structured ESG data and enrich metric relationships.

3) Map ESG indicators to ontology-based feature hierarchies for improved interpretability.

Phase 2: Principal Component Analysis (PCA) for ESG Data

This phase applies PCA to extract the most relevant ESG features while integrating ontology-based knowledge to enhance explainability.

Task 4: Applying PCA

1) Compute the covariance matrix and extract eigenvalues & eigenvectors.

2) Determine principal components (PCs) that retain the most variance in ESG data.

3) Reduce the dimensionality of ESG metrics, ensuring minimal information loss.

Task 5: PCA Interpretation & Feature Engineering

1) Visualize PCA results using scree plots, biplots, and heatmaps.

2) Interpret the significance of each principal component (PC) in ESG risk assessment.

3) Compare traditional PCA vs. ontology-enhanced PCA models for interpretability and feature selection.

Task 6: ESG Risk Factor Analysis

1) Identify which ESG categories contribute most to principal components.

2) Link PCA-extracted ESG factors to corporate sustainability risks and decision-making.

3) Validate findings using external ESG reports and industry benchmarks.

Phase 3: Model Implementation, Evaluation, and Reporting

This phase implements, evaluates, and presents the ontology-enhanced PCA model through an interactive dashboard and technical report.

Task 7: Model Validation and Comparison

1) Compare ontology-enhanced PCA with traditional PCA to assess model accuracy and interpretability.

2) Evaluate performance using variance explained, clustering accuracy, and ESG metric influence.

Task 8: ESG Dashboard

1) Develop a visual dashboard (Dash, Streamlit, or Tableau) to present key PCA-driven ESG insights.

Task 9: Final Presentation and Documentation

1) Prepare a presentation summarizing project outcomes and key findings.

2) Submit documented code, datasets, and results in a reproducible format.

Required Knowledge and skills:

The system will be developed as a data processing and ESG analytics platform, integrating ontology-based knowledge representation with PCA-driven feature selection and dimensionality reduction. The following features and specifications define the software/system requirements:

A. ESG Data Processing and Ontology Integration

- Feature Extraction

1) Load and preprocess Eurofidai ESG dataset from multiple reporting sources.

2) Handle missing data and ensure data consistency before PCA transformation.

- Ontology-Based ESG Categorization

1) Assign ESG metrics to predefined ontology-based categories (E, S, G).

2) Retrieve structured ESG relationships using SPARQL queries from the ontology knowledge base.

3) Link ESG metrics to industry-specific sustainability concepts.

- Data Transformation for PCA

1) Convert categorical ESG attributes into ontology-enriched numerical values.

2) Standardize ESG risk metrics for PCA processing.

B. PCA-Based ESG Feature Selection and Risk Assessment

- PCA Model Implementation

1) Compute eigenvalues, eigenvectors, and explained variance for principal component selection.

2) Apply dimensionality reduction while maintaining ESG interpretability.

- Visualization & Interpretability

1) Generate heatmaps, biplots, and component contribution charts for ESG insights.

2) Link PCA results to ontology concepts to enhance explainability.

3) Implement anomaly detection using PCA to identify ESG reporting inconsistencies.

C. ESG Dashboard

- Interactive Dashboard

1) Develop a real-time visualization tool for PCA-based ESG analytics.

2) Display principal components, risk scores, and ESG trend analysis.

Expected outcomes/deliverables:

1) Preprocessed ESG dataset with ontology-enhanced feature categorization.

2) PCA-based ESG factor analysis, identifying key sustainability risk drivers.

3) Ontology-driven feature engineering applied to ESG data.

4) Python-based implementation, including Jupyter Notebook documentation.

5) Comparative analysis of traditional PCA vs. ontology-enhanced PCA models.

6) Final project report & presentation, detailing methodology, findings, and business applications.

Project Number: 32

Project Title: Financial Market Analytics Application

Project Clients: Hirad Rezaei

Project specializations: Software Development;Web Application Development;Mobile Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;

Number of groups: 2 group

Main contact: Hirad Rezaei

Background:

Modern financial markets generate a wealth of data from diverse sources, including fundamental financial metrics and sentiment data from news, social media, and analyst reports. Investors require tools that not only consolidate this heterogeneous data but also analyze it to reveal actionable insights. Advances in AI—especially in LLMs—offer promising approaches to parsing the complex language of financial texts. This project challenges students to build an integrated application that automates data gathering, processing, and sophisticated analytics, while deploying the solution using Docker for ease of setup and reproducibility.

Project Goals:

• Develop an application that aggregates data from multiple financial APIs, collecting both fundamental metrics (e.g., earnings, ratios, balance sheets), sentiment data (news articles, social media feeds) and some technical indicators (like RSI, moving average)

• Design and implement a robust data pipeline to process, clean, and store the ingested data using a containerized database solution (e.g., Dockerized PostgreSQL or MongoDB).

• Perform in-depth analysis using machine learning techniques—emphasizing LLMs for domain-specific sentiment analysis —and explore other analytical models to gain insights into stock performance. (Fine-tuned LLMs are preferred)

• Create an interactive, user-friendly dashboard (web or mobile) to visualize trends, generate insights, and optionally trigger real-time alerts based on significant market changes.

• Package the entire application (data pipeline, ML models, and user interface) within Docker containers, using Docker Compose for orchestration to ensure easy deployment and portability.

Requirements and Scope:

• Develop an application that aggregates financial data—both fundamental metrics and sentiment insights—from multiple external APIs.

• Process and clean the collected data, then store it in a containerized database environment.

• Analyze the data using machine learning, with an emphasis on large language models for domain-specific analysis and incorporating other analytical models.

• Provide an interactive dashboard (web or mobile) that visualizes trends, displays insights, and may generate alerts for significant market changes.

• Deploy the entire solution using Docker with Docker Compose for ease of setup, reproducibility, and local/institutional deployment.

Required Knowledge and skills:

• API Integration:

– Connect to multiple financial data providers (e.g., Yahoo Finance, Alpha Vantage, Google News API) to collect both structured and unstructured financial data.

• Data Pipeline & Storage:

– Build an automated ETL pipeline to extract, transform, and load data into a Dockerized database solution (such as PostgreSQL or MongoDB).

– Ensure data integrity and security within the deployed environment.

• Machine Learning & Analysis:

– Fine-tune large language models on a curated corpus of financial texts for analysis.

– Evaluate these models against off-the-shelf alternatives and consider incorporating predictive analytics for stock trend forecasting.

• User Interface:

– Develop a responsive front-end (web or mobile) that displays analytics, trends, and alert mechanisms, with interactive features for user customization.

• Deployment & Infrastructure:

– Containerize all components of the application (API, database, ML modules, and user interface) using Docker.

– Provide a docker-compose configuration to orchestrate multi-container deployment on any system supporting Docker.

– Explore real-time data streaming (e.g., via Kafka) for live market tracking and alerts.

– Maintain a modular system design to facilitate the integration of additional data sources or future analytical tools.

Expected outcomes/deliverables:

• A fully functional Financial Market Analytics Application (accessible via web or mobile) that aggregates and processes real-time financial data.

• An automated and secure data pipeline integrated with multiple APIs and deployed using containerized services.

• An analysis module leveraging LLMs alongside other analytical models to provide actionable insights on different stocks.

• An interactive dashboard featuring visualizations, trend analysis, and alert mechanisms.

• Docker configuration (Dockerfiles and docker-compose file) ensuring that the entire application can be deployed and tested locally or on institutional infrastructure.

• Comprehensive documentation including a user guide, technical architecture overview, and source code repository.

Project Number: 61

Project Title: What's On! Student Life Passport

Project Clients: Rushi Vyas

Project specializations: Software Development;Web Application Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;Internet of Things (IoTs);Human Computer Interaction (HCI);

Number of groups: 4 groups

Main contact: Rushi Vyas

Background:

The "What's On! Student Life Passport" initiative aims to enhance student engagement and participation in campus activities. Recognizing that active involvement in university events fosters a sense of community, personal development, and overall student satisfaction, this project seeks to create a platform that encourages and rewards students for their engagement. By providing a centralized system to track participation and offer incentives, the project aspires to make campus life more vibrant and inclusive.

Project Goals:

1. Develop a User-Friendly Platform: Create an intuitive digital application where students can easily discover, register for, and check into campus events and activities.

2. Implement a Reward System: Establish a points-based system that tracks student participation, allowing them to earn rewards or recognition for their involvement in various events.

3. Enhance Event Visibility: Provide a comprehensive calendar and notification system to inform students about upcoming events, workshops, and activities, thereby increasing awareness and attendance.

4. Foster Community Engagement: Encourage students to explore diverse campus offerings, connect with peers, and build a stronger sense of belonging within the university community.

5. Collect Participation Data: Gather insights on student engagement patterns to help university organizers tailor future events and initiatives to better meet student interests and needs.

Requirements and Scope:

Project Scope:

The "What's On! Student Life Passport" project aims to design, develop, and deploy a comprehensive digital platform to enhance student engagement in campus activities at UNSW Sydney. The scope includes:

1. Platform Development:

Event Discovery: Implement a user-friendly interface that allows students to browse, search, and filter upcoming campus events across various categories such as academic, social, cultural, and recreational activities.

Registration and Check-In: Develop functionalities enabling students to register for events and check in upon arrival, utilizing methods like QR codes or geolocation services.

2. Reward System Implementation:

Points Accumulation: Design a system where students earn points for attending events, with varying points based on event type, size, or significance.

Incentives and Recognition: Create a tiered rewards structure offering incentives such as merchandise, certificates, or exclusive opportunities, and provide recognition through leaderboards or digital badges.

3. AI-Driven Skill Extraction:

Event Description Analysis: Utilize Natural Language Processing (NLP) techniques to analyze event descriptions, objectives, and activities.

Skill Identification: Employ AI models trained to recognize and extract both soft skills (e.g., communication, teamwork) and hard skills (e.g., programming, data analysis) associated with each event.

Student Skill Profiles: Automatically update individual student profiles with the skills acquired through event participation, providing a detailed overview of their skill development journey.

4. Communication Features:

Notifications: Set up a notification system to remind students of upcoming events they've registered for and inform them of new or recommended activities.

Feedback Mechanism: Incorporate a feature allowing students to provide feedback on events, aiding organizers in future planning.

5. Data Management and Analytics:

Participation Tracking: Ensure secure storage and management of student participation data, adhering to privacy policies.

Analytical Reporting: Develop tools for generating reports on engagement metrics to assist university staff in assessing and enhancing student involvement strategies.

6. Administrative Interface:

Event Management: Provide a backend interface for event organizers to create, update, and manage event listings.

Reward Management: Allow administrators to define reward criteria, manage inventory, and oversee the distribution process.

Exclusions:

The project will not cover the development of a mobile application; the platform will be web-based only.

Integration with external social media platforms is beyond the current scope.

The project will not handle financial transactions or payment processing for event registrations.

Assumptions:

Students have access to devices capable of accessing the web-based platform.

Event organizers are willing to adopt and regularly update the platform with their event information.

Adequate support from university IT services will be available throughout the project duration.

Required Knowledge and skills:

1. Platform Development

Event Discovery

Feature: Provide a searchable and filterable event calendar.
Enable event registration and attendance tracking.
Implement a points-based reward system.
Offer rewards and recognition for participation.
Analyze event content to identify associated skills.
Classify and record skills linked to events.
Maintain individual skill development records.
Send reminders and updates about events.
Collect student feedback on events.
Securely store and manage attendance data.
Generate reports on engagement metrics.

Allow organizers to manage event listings.
Oversee the rewards system.

Specification: Students can view events categorized by type (academic, social, cultural, recreational) and date. Filters for event type, date range, and keywords will be available.
Students can register for events and check in via QR codes or geolocation. The system records attendance and updates participation records.
Assign points to events based on criteria such as type, size, or significance. Points are automatically credited to students' profiles upon event check-in.
Develop a tiered rewards structure, including merchandise, certificates, or exclusive opportunities. Display leaderboards and digital badges to recognize active participants.
Utilize Natural Language Processing (NLP) to assess event descriptions and objectives, extracting keywords related to soft and hard skills.
Map extracted keywords to a predefined skills taxonomy, categorizing them into soft skills (e.g., communication, teamwork) and hard skills (e.g., programming, data analysis).
Automatically update students' profiles with skills acquired through event participation, providing a comprehensive view of their development.
Implement a system to notify students of upcoming events they've registered for and inform them of new or recommended activities based on their interests.
Provide a platform for students to rate and comment on events, offering insights for organizers to improve future activities.
Ensure compliance with data privacy policies in storing participation records, with access controls to protect student information.
Develop tools to analyze data on event attendance, participation trends, and skill development, aiding in strategic planning for student engagement.
Provide a backend interface for event organizers to create, update, and manage event details, including scheduling and capacity limits.
Enable administrators to define reward criteria, manage inventory, and monitor the distribution of incentives to students.

Registration and Check-In

2. Reward System Implementation

Points Accumulation

Incentives and Recognition

3. AI-Driven Skill Extraction

Event Description Analysis

Skill Identification

Student Skill Profiles

4. Communication Features

Notifications

Feedback Mechanism

5. Data Management and Analytics

Participation Tracking

Analytical Reporting

6. Administrative Interface

Event Management

Reward Management

Non-Functional Requirements

Performance

The system should handle concurrent access by up to 10,000 students without performance degradation.

Security

Implement authentication and authorization protocols to protect user data and ensure only authorized access to administrative features.

Usability

Design an intuitive user interface accessible across various devices and screen sizes, ensuring ease of navigation and interaction.

Scalability

Ensure the platform can scale to accommodate increasing numbers of users and events without compromising performance.

Compliance

Adhere to IT and data privacy policies, ensuring all data handling meets regulatory standards.

Expected outcomes/deliverables:

Upon successful completion of the project, the following outcomes and deliverables are anticipated:

1. Functional Web-Based Platform:

A fully operational web application that enables students to discover, register for, and check into campus events.

2. Source Code Repository:

Comprehensive and well-documented source code encompassing all components of the platform, including front-end, back-end, and database integrations.

Utilization of version control systems (e.g., Git) to manage code versions and facilitate collaboration.

3. Technical Documentation:

System Architecture Document: Detailed diagrams and explanations of the system's architecture, outlining components, interactions, and data flow.

API Documentation: Clear specifications of any Application Programming Interfaces (APIs) developed, including endpoints, request/response formats, and usage examples.

Database Schema: Comprehensive documentation of the database design, including entity-relationship diagrams and table structures.

4. User Documentation:

User Guide: A step-by-step manual guiding end-users through the platform's features and functionalities, enhanced with screenshots or illustrations for clarity.

Administrator Guide: Instructions tailored for system administrators and event organizers on managing events, users, and rewards within the platform.

5. AI Skill Extraction Module:

An integrated AI component that analyzes event descriptions to identify and categorize associated soft and hard skills, updating student profiles accordingly.

Documentation detailing the AI model's design, training process, and implementation specifics.

6. Testing Artifacts:

Test Plan: A comprehensive plan outlining the testing strategy, scope, objectives, resources, and schedule.

Test Cases: Detailed test scenarios covering various functionalities, including expected outcomes and execution steps.

Test Reports: Summaries of testing activities, results, identified defects, and their resolutions.

7. Project Management Documentation:

Project Plan: A document detailing the project's scope, objectives, timelines, milestones, and resource allocations.

Meeting Minutes: Records of discussions, decisions, and action items from project meetings.

Progress Reports: Regular updates tracking project status, accomplishments, challenges, and next steps.

8. Deployment Scripts and Instructions:

Automated scripts and detailed guidelines to facilitate the deployment of the platform in various environments (development, testing, production).

9. Final Presentation:

A comprehensive presentation summarizing the project's objectives, development process, challenges encountered, solutions implemented, and a live demonstration of the platform's capabilities.

Project Number: 62

Project Title: A Data-Driven Microtransit Platform Using Minibuses for Adaptive Urban Mobility Solutions

Project Clients: Ir. KK Leong & Dr. Matthew Cheng

Project specializations: Software Development;Web Application Development;Mobile Application Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;Cloud Computing;Computer Hardware and Networks;Security/Cyber Security;Internet of Things (IoTs);Human Computer Interaction (HCI);

Number of groups: 5 groups

Main contact: Ir. KK Leong & Dr. Matthew Cheng

Background:

As urban populations grow and mobility challenges intensify, the demand for efficient, sustainable, and user focused transportation solutions has never been greater. Traditional transit systems, often inflexible and outdated, fail to address critical issues such as traffic congestion, environmental pollution, and the need for

inclusive, adaptable services that cater to diverse communities. These limitations underscore the urgent need

for innovative solutions that can adapt to the evolving demands of modern urban life while ensuring accessibility

and convenience for all.

This project addresses these challenges by developing a data-driven microtransit platform that leverages fully

customizable 16-seater, 22-seater, and 30-seater minibuses as a core mobility solution tailored to the unique

transportation needs of the Malaysian and Singaporean markets. These minibuses are designed to bridge two

critical gaps in conventional transportation: wheelchair accessibility and ample luggage storage. Unlike

traditional taxis or modern e-hailing services that rely on standard passenger vehicles or SUVs, these minibuses can be outfitted with specialized features like wheelchair ramps, securement systems, and priority seating,

ensuring safe and comfortable travel for individuals with disabilities or limited mobility. Furthermore, they offer

spacious luggage compartments to accommodate passengers with baggage, such as tourists, airport travellers,

or students - a convenience often lacking in conventional transit options. With their optimal balance of capacity,

flexibility, and efficiency, these minibuses are ideal for a wide range of urban and suburban transit needs.

Inspired by industry players like TransLoc, Moovit, Via, and ReachNow, the platform will integrate cutting-edge

technologies to optimize transit systems, reduce environmental impact, and enhance urban mobility. By offering

on-demand, shared, and efficient ride solutions, the platform will alleviate traffic congestion and promote

sustainable mobility across key sectors, including hotels and resorts, theme parks, paratransit, student transit,

corporate and campus shuttle services, and health and aged care transportation.

The customizable minibuses ensure scalability and adaptability, enabling the platform to serve both small

communities and high-demand urban areas effectively. Designed to meet the diverse needs of modern travellers,

these vehicles combine inclusivity, convenience, and sustainability.

At its core, this project emphasizes the integration of these minibuses as a versatile, scalable, and inclusive mobility solution, supported by data-driven optimization, sustainability, and profitability. By combining advanced technology with a focus on inclusivity, convenience, and environmental responsibility, this initiative seeks to redefine urban transportation for the future.

Requirements and Scope:

Develop a scalable, cloud-based microtransit platform that leverages real-time data, artificial intelligence (AI), and advanced analytics to enhance shared mobility solutions. The platform will utilize 16, 22, and 30-seater minibuses to serve diverse sectors, including hotels, resorts, theme parks, airports, healthcare and aged care transportation, university campuses and student transit, corporate and campus shuttle services, and paratransit services. This innovative system aims to optimize operations, improve efficiency, and deliver seamless transportation experiences across various industries.

Required Knowledge and skills:

Technology Stack:

• Frontend: Develop an intuitive mobile app (iOS/Android) for booking rides, tracking vehicles, and accessing real-time traffic and route updates.

• Backend: Build a scalable, cloud-based infrastructure to manage real-time data, ride-sharing, and vehicle routing.

• AI & Machine Learning: Implement advanced algorithms for route optimization, traffic prediction, and dynamic service adjustments.

• Passenger Display Systems: Design in-vehicle screens to provide real-time transit information.

• Payment Integration: Support seamless payment processing for passengers and operators.

Expected outcomes/deliverables:

• Source Code

• Documentation

• User Guide

• Presentation deck

Project Number: 63

Project Title: SAT Decisions Scraper & Search Application with RAG (Retrieval-Augmented Generation)

Project Clients: Josh Tedjasaputra (Customer Experience Insight Pty Ltd)

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Human Computer Interaction (HCI);Internet of Things (IoTs);Cloud Computing;Big data Analytics and Visualization;Computer Science and Algorithms;Mobile Application Development;

Number of groups: 4 groups

Main contact: Josh Tedjasaputra (Customer Experience Insight Pty Ltd)

Background:

The project aims to develop an agentic Python application that automatically scrapes, processes, and indexes SAT (State Administrative Tribunal) decisions. The application should allow natural language queries using Retrieval-Augmented Generation (RAG) via AI Language Model to provide contextually relevant results, summaries, and insights. Additionally, the system must allow authorised users to manually upload new SAT decision documents, which will be processed, embedded, and immediately searchable. The primary goal is to enable efficient search and analysis of tribunal decisions using AI-driven insights (democratisation of law in WA)

Requirements and Scope:

- Develop a web-based search and retrieval system for SAT decisions.

- Implement an automated web scraper to extract tribunal decisions, metadata, and summaries.

- Integrate Retrieval-Augmented Generation (RAG) using AI Model to generate contextualised summaries of decisions.

- Provide a secure user interface for natural language queries and decision retrieval.

- Enable manual document addition with real-time indexing and searchability.

- Implement error handling, logging, and robust documentation for maintainability.

Required Knowledge and skills:

Data Acquisition & Processing:

- Scrape and store SAT decisions with metadata (date, case title, involved parties, summary).

- Handle structured and unstructured data cleaning (removal of duplicates, normalisation).

- Store the processed decisions in a searchable vector database (e.g., FAISS, Pinecone).

Search & Retrieval Interface:

- Web-based UI (Flask/FastAPI) allowing natural language queries (e.g., "Show decisions about environmental regulations").

- Query results should include decision links, metadata, and AI Model-generated summaries.

RAG & AI Integration:

- Use AI model embeddings for vector-based search and response generation.

- Implement context-aware retrieval of case details and similar decisions.

Manual Document Addition:

- Allow authenticated users to upload new decisions (PDF/DOCX).

- Process uploaded documents, generate embeddings, and make them searchable instantly.

System Scalability & Security:

- The system should run on a Linux-based environment with secured API endpoints.

- User authentication for manual document upload and administrative controls.

Expected outcomes/deliverables:

Functional Web Application that allows SAT decision search and retrieval.

Automated Scraper that collects SAT decisions and stores them in a structured format.

AI-Powered Search Engine using RAG with Llama embeddings.

Manual Document Addition Feature with real-time indexing.

Comprehensive Documentation (README, API instructions, user guide).

Error Handling & Logging Framework for system maintainability.

Project Number: 64

Project Title: What's On! Student Passport

Project Clients: What's On! Campus Pty Ltd

Project specializations: Web Application Development;Data Analytics & Visualization;

Number of groups: 3 groups

Main contact: What's On! Campus Pty Ltd

Background:

The "What's On! Student Life Passport" initiative aims to enhance student engagement and participation in campus activities on campus at a University. Recognizing that active involvement in university events fosters a sense of community, personal development, and overall student satisfaction, this project seeks to create a platform that encourages and rewards students for their engagement. By providing a centralized system to track participation and offer incentives, the project aspires to make campus life more vibrant and inclusive.

Project Goals:

1. Develop a User-Friendly Platform: Create an intuitive digital application where students can easily track past events.

2. Implement a Reward System: Establish a points-based system that tracks student participation, allowing them to earn rewards or recognition for their involvement in various events.

3. Foster Community Engagement: Encourage students to explore diverse campus offerings, connect with peers, and build a stronger sense of belonging within the university community.

Requirements and Scope:

Features:

History Records:

Functionality: Fetch data of each student's attendance and participation in campus events.
Provide personalized insights and support to students based on their event participation history.
Retrieve and store data on student attendance and participation in various events.
Analyze student engagement data to assign a reflecting their level of involvement.
Manage user authentication processes, ensuring secure access to the platform.

Purpose: Enable students to review their engagement history and allow the system to analyze participation patterns for personalized feedback.
Act as a virtual career tracker, offering insights on past extracurricular activities that align with students' interests and career aspirations.
Provide accurate records for analysis.
Identify areas where students can improve their engagement and provide insights.
Protect user data and provide personalized experiences based on user profiles.

Insights & Analytics Assistant:

Required Knowledge and skills:

Front-end Components:

Landing Page

Login/Registration:

Description: Secure authentication system for users to access personalized features.
A dashboard status indicator to assess and display students' engagement insights.

User registration with credentials.

Login portal for returning users.

Password recovery options.

Insights & Analysis from Student Attendance History as a Career Coach:

Visual representation of participation metrics.

Personalized insights on past activities for career prospects.

Back-end Components:

Get Event History:

Insights & Analysis Based on Profile:

Authentication (Login/Register):

Expected outcomes/deliverables:

Source Code, Documentation, User Guide

Project Number: 65

Project Title: My Pressure Story: A framework to help people deal with pressure

Project Clients: Nick Patrikeos

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Cloud Computing;

Number of groups: 1 group

Main contact: Nick Patrikeos

Background:

The Growth Hunting Company (https://www.growthhunting.com.au) is a startup dedicated to fostering growth and development of individuals and teams. They offer several frameworks, including:

• "Growth Zone Goal Setting" for goal assistance.

• "My Pressure Story™" for understanding pressure triggers and coping strategies.

"My Pressure Story™" is designed to help individuals develop self-awareness regarding pressure and offers strategies to manage it.

Video introduction: https://www.youtube.com/watch?v=Renr5MV79gA

Original Spec: https://drive.google.com/file/d/18yRnMd0P8fWNpMZ9-Z6oqAnlXhWA3LZB/view?usp=sharing

Requirements and Scope:

This project involves building on the existing MVP to deliver a production-ready version for the Growth Hunting website. Scope includes:

• Setting up a payment gateway for service access.

• Developing a “freemium” model with limited features.

• Enhancing data analytics capabilities for user insights.

• Implementing internationalisation for global audiences.

• Strengthening software robustness and quality assurance.

Required Knowledge and skills:

Participants will work on a repository comprising:

• React frontend

• TypeScript backend

Essential skills:

• TypeScript and React development

• UI/UX design principles

• Product requirements engineering

• Experience with internationalisation frameworks

Expected outcomes/deliverables:

Deliverables for this project:

• A fully functional production version of "My Pressure Story™" with integrated payment gateway and analytics.

• Technical documentation, user guide, and a well-commented codebase.

• Internationalisation support for multiple languages.

• Deployment on the Growth Hunting website with a freemium model.

• A system demo showcasing the core functionalities.

Project Number: 66

Project Title: Atlassian Forge

Project Clients: Khanh Nguyen

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Cloud Computing;

Number of groups: 1 group

Main contact: Khanh Nguyen

Background:

Forge is an Atlassian Platform that allows developers to build fully-functional apps rapidly, with built-in hosting, multiple development environments, and API authentication. It enables the creation of custom apps and integrations for Atlassian products as well as apps distributed via the Atlassian Marketplace. For more details, students are encouraged to explore the official Forge documentation.

Please see here for more info:
https://drive.google.com/file/d/107MQbWRB3zNIFpQU5fu9Ymj8K6exRmMN/view?usp=sharing

Requirements and Scope:

Students can choose from one of several Forge app project ideas or propose their own. The project options include:
• AI-Powered: Create a slide deck from a Confluence page (with AI-generated speaker notes and visuals).
• AI-Powered: Draft a Confluence page from a voice memo.
• Confluence whiteboard to Confluence page: Automate the transformation of whiteboard content into a Confluence page.
• Team Tamagotchi: Develop a virtual team pet accessible across Jira and Confluence.
• Confluence note taker: Create an app to take notes as you read a Confluence page.
• Dietaries tracker: Build an app to persist dietary information for users, aiding event organization.

The project involves developing a Forge app using Atlassian's integrated hosting and API authentication features. Students are expected to deliver a working app based on one of the project briefs.

Required Knowledge and skills:

• Proficiency in JavaScript programming (COMP1531 knowledge is sufficient).
• Basic React skills are helpful (and can be self-taught if needed).
• Familiarity with the Atlassian Forge platform and its development environment will be beneficial.

Expected outcomes/deliverables:

A fully functional Forge app deployed on the Atlassian platform.
• Well-documented source code and technical documentation.
• A live demonstration of the app's core functionalities.
• A report outlining the design process, development challenges, and solution implementation.
• A user guide and testing artifacts.

Project Number: 67

Project Title: Atlassian Forge

Project Clients: Jie Mee Chong

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Cloud Computing;

Number of groups: 1 group

Main contact: Jie Mee Chong

Background:

Forge is an Atlassian Platform that allows developers to build fully-functional apps rapidly, with built-in hosting, multiple development environments, and API authentication. It enables the creation of custom apps and integrations for Atlassian products as well as apps distributed via the Atlassian Marketplace. For more details, students are encouraged to explore the official Forge documentation.

Please see here for more info:
https://drive.google.com/file/d/107MQbWRB3zNIFpQU5fu9Ymj8K6exRmMN/view?usp=sharing

Requirements and Scope:

Students can choose from one of several Forge app project ideas or propose their own. The project options include:
• AI-Powered: Create a slide deck from a Confluence page (with AI-generated speaker notes and visuals).
• AI-Powered: Draft a Confluence page from a voice memo.
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The project involves developing a Forge app using Atlassian's integrated hosting and API authentication features. Students are expected to deliver a working app based on one of the project briefs.

Required Knowledge and skills:

• Proficiency in JavaScript programming (COMP1531 knowledge is sufficient).
• Basic React skills are helpful (and can be self-taught if needed).
• Familiarity with the Atlassian Forge platform and its development environment will be beneficial.

Expected outcomes/deliverables:

A fully functional Forge app deployed on the Atlassian platform.
• Well-documented source code and technical documentation.
• A live demonstration of the app's core functionalities.
• A report outlining the design process, development challenges, and solution implementation.
• A user guide and testing artifacts.

Project Number: 68

Project Title: Atlassian Forge

Project Clients: Cheryl Leung

Project specializations: Software Development;Web Application Development;Cloud Computing;Artificial Intelligence (Machine/Deep Learning, NLP);

Number of groups: 1 group

Main contact: Cheryl Leung

Background:

Forge is an Atlassian Platform that allows developers to build fully-functional apps rapidly, with built-in hosting, multiple development environments, and API authentication. It enables the creation of custom apps and integrations for Atlassian products as well as apps distributed via the Atlassian Marketplace. For more details, students are encouraged to explore the official Forge documentation.

Please see here for more info:
https://drive.google.com/file/d/107MQbWRB3zNIFpQU5fu9Ymj8K6exRmMN/view?usp=sharing

Requirements and Scope:

Students can choose from one of several Forge app project ideas or propose their own. The project options include:
• AI-Powered: Create a slide deck from a Confluence page (with AI-generated speaker notes and visuals).
• AI-Powered: Draft a Confluence page from a voice memo.
• Confluence whiteboard to Confluence page: Automate the transformation of whiteboard content into a Confluence page.
• Team Tamagotchi: Develop a virtual team pet accessible across Jira and Confluence.
• Confluence note taker: Create an app to take notes as you read a Confluence page.
• Dietaries tracker: Build an app to persist dietary information for users, aiding event organization.

The project involves developing a Forge app using Atlassian's integrated hosting and API authentication features. Students are expected to deliver a working app based on one of the project briefs.

Required Knowledge and skills:

• Proficiency in JavaScript programming (COMP1531 knowledge is sufficient).
• Basic React skills are helpful (and can be self-taught if needed).
• Familiarity with the Atlassian Forge platform and its development environment will be beneficial.

Expected outcomes/deliverables:

A fully functional Forge app deployed on the Atlassian platform.
• Well-documented source code and technical documentation.
• A live demonstration of the app's core functionalities.
• A report outlining the design process, development challenges, and solution implementation.
• A user guide and testing artifacts.

Project Number: 69

Project Title: Atlassian Forge

Project Clients: Ajay Mathur

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Cloud Computing;

Number of groups: 1 group

Main contact: Ajay Mathur

Background:

Forge is an Atlassian Platform that allows developers to build fully-functional apps rapidly, with built-in hosting, multiple development environments, and API authentication. It enables the creation of custom apps and integrations for Atlassian products as well as apps distributed via the Atlassian Marketplace. For more details, students are encouraged to explore the official Forge documentation.

Please see here for more info:
https://drive.google.com/file/d/107MQbWRB3zNIFpQU5fu9Ymj8K6exRmMN/view?usp=sharing

Requirements and Scope:

Students can choose from one of several Forge app project ideas or propose their own. The project options include:
• AI-Powered: Create a slide deck from a Confluence page (with AI-generated speaker notes and visuals).
• AI-Powered: Draft a Confluence page from a voice memo.
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• Confluence note taker: Create an app to take notes as you read a Confluence page.
• Dietaries tracker: Build an app to persist dietary information for users, aiding event organization.

The project involves developing a Forge app using Atlassian's integrated hosting and API authentication features. Students are expected to deliver a working app based on one of the project briefs.

Required Knowledge and skills:

• Proficiency in JavaScript programming (COMP1531 knowledge is sufficient).
• Basic React skills are helpful (and can be self-taught if needed).
• Familiarity with the Atlassian Forge platform and its development environment will be beneficial.

Expected outcomes/deliverables:

A fully functional Forge app deployed on the Atlassian platform.
• Well-documented source code and technical documentation.
• A live demonstration of the app's core functionalities.
• A report outlining the design process, development challenges, and solution implementation.
• A user guide and testing artifacts.

Project Number: 70

Project Title: Vehicle Parts Recommender

Project Clients: Kim Horn

Project specializations: Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Computer Science and Algorithms;Big data Analytics and Visualization;

Number of groups: 3 groups

Main contact: Kim Horn

Background:

The aim of this project is to build a product recommender, using past data. Market basket analysis is one of the most used types of Machine Learning in the business space. Infomedia is a global Parts Data company: infomedia.com.au. Infomedia deals with Automotive manufacturers and Vehicle dealerships and processes their data. One of Infomedia's main products is a B2B and B2C SaaS Vehicle Part ordering system used by many auto dealerships. The goal is to add Market Basket analysis to the product.

Requirements and Scope:

From a history of Vehicle Parts Orders build a market basket model that can recommend 'relevant associated' product (part) items when a customer is adding single part items to their order basket. The data provided will include a large set of vehicle parts orders, and a graph of how these parts are related. The delivered system will be in two parts:

1. It will input a file of parts data, process the data, and a machine learning system will produce a recommender model. The model may be augmented with domain rules. Provide a viewer for Product users to see and edit the model rules.

2. The model will be input to a front end GUI Assistant System, that takes in user part items (by ID), in a basket, and generates recommendations. This may be two modules, a front web GUI, and a backend REST API.

Required Knowledge and skills:

Explore and understand market basket technical approaches available today, e.g. Apriori, Eclat, FPGrowth, and document pros and cons. Choose the best approach and use the data to build a model. Provide a way for domain experts, mechanics and Parts staff, to view and understand the rules (model) generated. Explore and research how to represent the rules explicitly in, for example a graph, and then how to mix in background and domain knowledge, that provides a deeper meaning for 'associated' parts. Parts may be connected physically, replaced together etc. Make part orders goal oriented rather than simply a list.

For example, a Dealership needs to fix a customer's problem, and the order can be optimised for this repair by ordering more parts to solve common issues, but knowing they all may not be required. Having more parts means the time to fix is reduced. This kind of 'information' could be added to the graph. Provide a user interface so the Infomedia Product team can view, edit and understand the rules generated. Provide a user interface so customers can order items and get recommendation for other parts interactively. More detailed use cases will be provided in the project, and a more definitive spec.

Expected outcomes/deliverables:

The outcome is two modules, a model generator and the assistant front end. Infomedia will provide all background information, expert knowledge, in both the domain and can also in Market basket ML approaches, Mechanical domain and background knowledge, and rules, about parts will be provided, with demonstrations of the parts ordering system.

Project Number: 71

Project Title: Code Solving Web Platform

Project Clients: Syed Mahbubul Huq

Project specializations: Software Development;Web Application Development;Cloud Computing;

Number of groups: 2 group

Main contact: Syed Mahbubul Huq

Background:

Coding platforms like LeetCode, HackerRank, and Codewars are platforms to help students develop their problem-solving skills. These platforms provide coding challenges that help users improve their knowledge in programming.

This project aims to create a problem-solving platform similar to LeetCode, focusing on Python as the language for problem-solving. The goal is to allow users to log in and solve coding problems and get feedback in an interactive coding environment. The web application will be built using a microservices architecture.

Requirements and Scope:

The project scope will cover the development of a coding problem-solving platform focused on Python. It will provide key functionalities such as problem categorization, login management, and an interactive coding IDE for logged-in users. Non-logged-in users will only be able to browse problems but will not be able to interact with them. In the current scope the user writes Python code to solve a problem and submits it through the platform. The backend receives the submission, executes the code in an isolated Docker container, and fetches the relevant test cases from MongoDB. The code is then run against the test cases, and the results are compared to determine whether the solution is correct.

The specific scope includes:

User Authentication & Management

Problem Categorization & Management

Coding Environment (IDE)

Non-logged-in User Interaction

Leaderboard (Basic)

Admin Panel

Required Knowledge and skills:

User Authentication & Profile Management

User Registration and Login: Users can sign up by providing email, username, and password, or log in to an existing account.

Session Management: Users will remain logged in unless they manually log out.

Profile Management: Users can update their personal profile, including changing their display name, profile image, and adding a bio.

Problem Management

Problem Categories: Problems will be divided into three categories: Easy, Medium, and Hard.

Problem Creation: Admins can create new problems with a problem statement, expected outputs, and test cases.

Problem Solving: Logged-in users will be able to attempt solving the problems and test their solutions.

Problem Status: Users can mark problems as solved upon successful execution of their code.

IDE Integration for Coding

Python IDE: A real-time Python coding environment will be available for solving problems.

Code Execution: Users can run their code and receive feedback on whether their solution is correct, along with any error messages or outputs.

Non-logged-in Users

Problem Viewing: Non-logged-in users can view the problems, their descriptions, and their categories but cannot interact with the IDE.

Leaderboard

Display Leaderboard: A leaderboard will show the top users based on the number of problems solved (logged-in users only).

Admin Panel

User Management: Admins can view and manage user data.

Problem Management: Admins can manage problems, including creating new ones or deleting existing ones.

Mobile/Desktop Responsiveness

The platform should work seamlessly across both mobile devices and desktop computers.

Scalability

The platform should be scalable to support an increasing number of users and problems.

The backend should handle a growing amount of users simultaneously without performance degradation.

Performance

Code execution should be quick, with results appearing within a reasonable timeframe.

The system should handle concurrent users efficiently.

Security

Data Encryption: All sensitive user data will be securely encrypted (e.g., passwords).

Authentication: Use OAuth or JWT tokens to ensure secure, stateless authentication.

Data Privacy: Users' personal information and solution data will be stored securely and in compliance with regulations.

Expected outcomes/deliverables:

Complete project documentation; documented source code, requirements, design, software architecture, testing, technology stack and technical dependcies/libraries/frameworks, automated deployment, user manual and deployment guide

Project Number: 72

Project Title: Video Sharing Web Platform

Project Clients: Syed Mahbubul Huq

Project specializations: Software Development;Web Application Development;Cloud Computing;

Number of groups: 2 group

Main contact: Syed Mahbubul Huq

Background:

Video-sharing platforms like YouTube, Vimeo, and Dailymotion have gained popularity in this digital era. On these platforms, video creators share their content with audiences, providing information, education, or entertainment. These platforms offer a smooth experience for both users and video uploaders.

This project aims to create a video-sharing platform that will serve as the foundation for user interactions involving video uploads, sharing, viewing, and social interactions. The platform will feature a basic set of functionalities that mirror some of the key aspects of existing platforms like YouTube while also being flexible enough for future expansion by following microservice architecture. The platform will allow registered users to upload videos, manage their profiles, and engage with content. It also allows non-logged-in users to freely view content.

Requirements and Scope:

The project's scope will cover the creation of the video-sharing platform with essential components of popular video-sharing platforms like YouTube. The scope of the project is but not limited to:

User Authentication & Management

Video Uploading & Management

Video Playback

Search & Discovery

User Interaction & Social Features

Admin Panel

Mobile and Desktop Responsiveness

Required Knowledge and skills:

User Authentication & Profile Management

User Registration and login: Users can create an account by providing an email address, username, and password.

Login/Logout: Users can log in using their registered email and password. Support for session management.

Password Management: Forgot password functionality, secure password reset through email.

Profile Management: Users can update their profile, including their username, profile picture, and bio.

Video Uploading

File Upload: Users can upload video files in popular formats (MP4, AVI, MOV).

Metadata: Each uploaded video must include metadata (title, description, tags, and a thumbnail image).

Video Processing: Videos should be automatically transcoded into multiple formats/resolutions to optimize for different devices (1080p, 720p, 480p).

Progress Bar: A progress bar will show the status of video upload and processing.

Video Viewing

Playback Controls: Users can play/pause, skip, adjust the volume, and toggle full-screen mode.

Streaming: Videos should stream smoothly, with buffering only occurring when absolutely necessary.

Video Information: Display video statistics like view count, like/dislike ratio, and the number of comments.

Search & Discovery

Search: Users should be able to search for videos using keywords, titles, or tags.

Sorting: Provide users with options to sort videos by most recent, most popular, or highest-rated.

User Interactions (Social Features)

Likes & Dislikes: Users can express feedback on videos by liking or disliking them.

Comments: Users can leave comments under videos. The ability to upvote, downvote, or report comments for inappropriate content.

Subscriptions: Registered users can subscribe to content creators, enabling them to receive notifications about new uploads.

Sharing: Users should be able to share videos via social media or direct links.

Admin Panel

Dashboard: Admins can view the platform's statistics, such as active users, recent uploads, and reports.

User Management: Admins can ban any users who violate terms of service.

Content Management: Admins can remove inappropriate comments.

Mobile/Desktop Responsiveness:

The design will be fully responsive, ensuring it works seamlessly across mobile devices (iOS/Android) and desktop computers.

Scalability

The platform should handle a growing user base and video content.

The backend should be capable of scaling horizontally (adding more servers or microservices as needed).

Media storage (videos and images) should scale using cloud storage solutions (AWS S3 or Google Cloud Storage).

Performance

Video uploads should have minimal wait times, and video playback should begin quickly.

The system sho

Expected outcomes/deliverables:

Complete project documentation; documented source code, requirements, design, software architecture, testing, technology stack and technical dependcies/libraries/frameworks, automated deployment, user manual and deployment guide